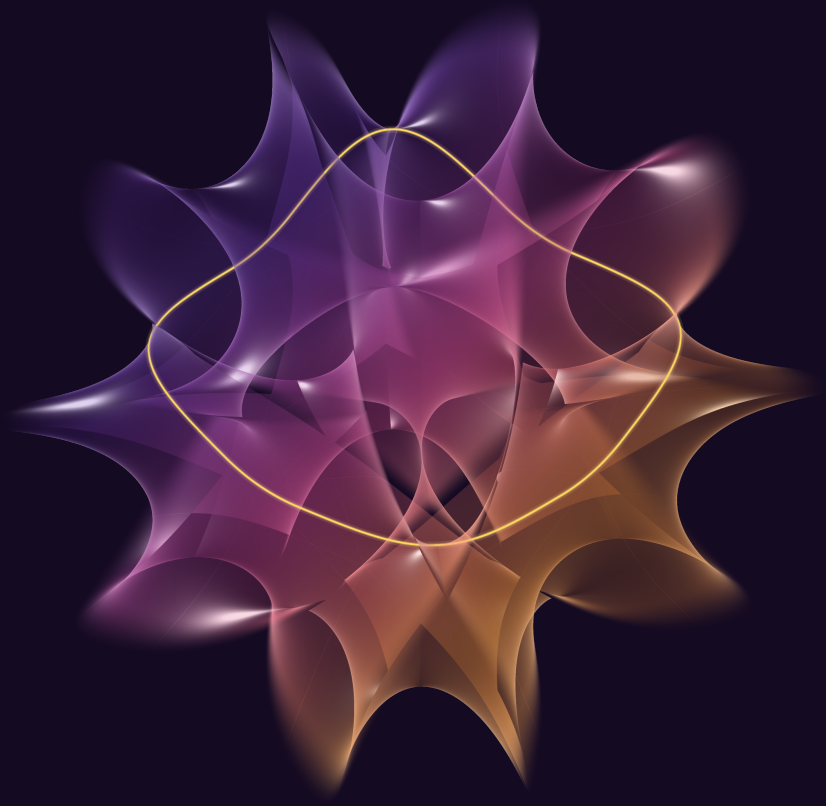


Introduction to String Theory

Part II: String compactifications

Lecture notes on the module of Prof. Michele Cicoli
University of Bologna, academic year 2026/27

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Notation and conventions

This chapter fixes one set of conventions for the whole of the notes. They are the ones most common in the string-compactification literature, which are not always those used on the board; where a lecture writes a formula in another convention, it is converted to the one below.

Units and approximate equalities. We use natural units, $\hbar = c = 1$, so that masses, energies and inverse lengths are measured in GeV. An equality sign is exact. The sign "$\simeq$" means equality up to numerical factors of order one (such as 2, π or $\sqrt{4\pi}$) that are dropped when only the parametric dependence on g_s and $\mathcal{V}$ matters, and "$\sim$" means "of the order of". When a relation is first derived its factors are kept, and it is then quoted with "$\simeq$".

Metric and curvature. Metrics have mostly-plus signature, $\eta_{MN} = \mathrm{diag}(-1, +1, \ldots, +1)$, as in Polchinski [7], Tong [8], Wess and Bagger, and most of the compactification literature. The Riemann tensor is $R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}$, the Ricci tensor is $R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$, and a sphere has positive Ricci scalar. With these choices the four-dimensional Einstein–Hilbert action is

$$S_{\mathrm{EH}} = \frac{M_P^2}{2} \int \mathrm{d}^4x \, \sqrt{-g}\, R .$$

For a Lorentzian metric we write $\sqrt{-g}$, for the Riemannian metric of the compact space $\sqrt{g}$.

Spaces, coordinates and indices. Ten-dimensional spacetime is $X_{10} = \mathbb{R}^{1,3} \times Y_6$, with Y_6 a compact six-dimensional manifold and coordinates $X^M = (x^\mu, y^m)$.

- Capital Latin indices $M, N, P = 0, \ldots, 9$ are ten-dimensional.
- Greek indices $\mu, \nu, \rho = 0, \ldots, 3$ are four-dimensional.
- Lower-case Latin indices $m, n, p = 4, \ldots, 9$ are real indices on Y_6.
- The indices $i, j = 1, 2, 3$ and $\bar{\imath}, \bar{\jmath}$ label holomorphic and antiholomorphic coordinates z^i, $\bar{z}^{\bar{\jmath}}$ on Y_6 when it is a complex manifold.

The ten-dimensional metric is G_{MN}, with determinant G and Ricci scalar R_{10}. On a product space it splits into the four-dimensional metric $g_{\mu\nu}$ (determinant g_4, Ricci scalar R_4) and the internal metric g_{mn} (determinant g_6, Ricci scalar R_6). The volume of the compact space is $\mathrm{Vol}(Y_6) = \int_{Y_6} \mathrm{d}^6y \, \sqrt{g_6}$. The holonomy group of Y_6 is $\mathrm{Hol}(Y_6)$, and $\mathbb{1}$ is the identity.

String length and string scale. The only dimensionful parameter of string theory is α', related to the string tension by $T = 1/(2\pi\alpha')$. Following the lecture and the moduli-stabilisation literature we define

$$\ell_s = 2\pi\sqrt{\alpha'}, \qquad M_s = \frac{1}{\ell_s}.$$

Polchinski and Tong use $\ell_s = \sqrt{\alpha'}$ instead, which is smaller by 2π. The virtue of the factor 2π is that it absorbs the powers of 2π of the ten-dimensional gravitational coupling, $2\kappa_{10}^2 = (2\pi)^7\alpha'^4 = \ell_s^8/2\pi$, and of the D$p$-brane tensions, $T_p = 2\pi M_s^{p+1}/g_s$. The price is that T-duality, which exchanges a circle of radius R with one of radius α'/R, reads $R \to \alpha'/R = 1/(4\pi^2 M_s^2 R)$ (Equation (2.3)); the self-dual radius is $\sqrt{\alpha'} = \ell_s/2\pi$.

Dilaton, string coupling and frames. The dilaton ϕ is a field. Its vacuum expectation value is $\phi_0 \equiv \langle\phi\rangle$, and the string coupling is

$$g_s = e^{\phi_0}.$$

The *string frame* metric G_{MN} is the one to which the string couples on the worldsheet; in it the gravitational action carries a factor $e^{-2\phi}$ in front of the Ricci scalar (Equation (1.5)). The ten-dimensional *Einstein frame* metric is

$$G^E_{MN} = e^{-(\phi-\phi_0)/2}\, G_{MN},$$

in which the gravitational action takes the Einstein–Hilbert form $\frac{1}{2\kappa_{10}^2 g_s^2}\int \mathrm{d}^{10}X\sqrt{-G_E}\,R_{10}[G_E]$. Because ϕ_0 is subtracted, the two frames coincide in a vacuum with constant dilaton, $\phi = \phi_0$. In particular a volume measured in string units is the same number in both frames, and the relations between scales derived in Chapter 1, $M_P^2 = 4\pi\mathcal{V}M_s^2/g_s^2$ and $M_{KK} \simeq M_s/\mathcal{V}^{1/6}$, hold in either.

Much of the moduli-stabilisation literature uses instead $G^E_{MN} = e^{-\phi/2}G_{MN}$, without the subtraction. In a vacuum the two Einstein frames then differ by the constant factor $g_s^{-1/2}$, so that a dimensionless volume in that convention is $\mathcal{V}' = g_s^{-3/2}\,\mathcal{V}$, and the same relations read $M_s = g_s^{1/4}M_P/\sqrt{4\pi\mathcal{V}'}$ and $M_{KK} \simeq M_P/(\sqrt{4\pi}\,\mathcal{V}'^{2/3})$. Formulas taken from such references are converted with $\mathcal{V}' = g_s^{-3/2}\mathcal{V}$.

Volume. $\mathcal{V}$ is the dimensionless volume of Y_6 in units of ℓ_s^6, measured with the Einstein-frame metric:

$$\mathrm{Vol}(Y_6) = \mathcal{V}\,\ell_s^6.$$

Planck mass. M_P is the reduced Planck mass,

$$M_P = (8\pi G_N)^{-1/2} \simeq 2.4\times10^{18}\,\mathrm{GeV},$$

the coefficient of the four-dimensional Einstein–Hilbert action above. Four-dimensional actions are written in the four-dimensional Einstein frame, in which this coefficient is constant.

Counting supersymmetries. $\mathcal{N}$ counts supersymmetry generators in units of the smallest spinor of the spacetime dimension. In ten dimensions this is a Majorana–Weyl spinor, with 16 real components, so type IIA and IIB supergravity, with two of them, have ten-dimensional $\mathcal{N} = 2$ and 32 real supercharges. The two have opposite chirality in type IIA, and the same chirality in type IIB, which is therefore chiral. In four dimensions the unit is a Majorana spinor, equivalently a Weyl spinor and its conjugate, with 4 real components, so that four-dimensional $\mathcal{N} = 8, 4, 2, 1$ correspond to $32, 16, 8, 4$ real supercharges. A subscript $\pm$ on a ten-dimensional spinor, as in ψ_{M+} or λ_-, gives its chirality, the eigenvalue ±1 of $\Gamma_{11} = \Gamma^0\Gamma^1\cdots\Gamma^9$, which squares to one in our signature.

Forms. A p-form $C_p = \frac{1}{p!} C_{M_1 \ldots M_p}\, \mathrm{d}X^{M_1} \wedge \cdots \wedge \mathrm{d}X^{M_p}$ is identified with its totally antisymmetric components, e.g. $C_2 \leftrightarrow C_{MN} = C_{[MN]}$. The NS–NS two-form is B, with field strength $H = \mathrm{d}B$; the R–R forms are C_p, with field strengths $F_{p+1} = \mathrm{d}C_p + \ldots$. A form ω on Y_6 is harmonic if $\Delta\omega = 0$, with $\Delta = \mathrm{d}\mathrm{d}^\dagger + \mathrm{d}^\dagger\mathrm{d}$ the Laplacian of the internal metric. On a complex manifold a (p, q)-form has p holomorphic and q antiholomorphic indices.

Moduli. The Hodge numbers $h^{p,q}$ of Y_6 count its linearly independent harmonic (p, q)-forms; for a Calabi–Yau threefold the non-trivial ones are $h^{1,1}$ and $h^{2,1}$ ($= h^{1,2}$). A generic modulus is ϕ^I; in discussing the couplings of moduli to matter, ϕ without index also stands for a generic canonically normalised modulus. The Kähler moduli are T_i, $i = 1, \ldots, h^{1,1}$, and the complex-structure moduli U_a, $a = 1, \ldots, h^{2,1}$. The index i of T_i is a label, not a holomorphic index; the two are never needed in the same formula, and the context tells which is meant. The axio-dilaton is

$$S = e^{-\phi} + i\, C_0, \qquad \langle \mathrm{Re}\, S \rangle = \frac{1}{g_s},$$

as in the lecture and in the Large Volume Scenario literature. Much of the flux-compactification literature (GKP, KKLT, F-theory) uses instead $\tau = C_0 + i\, e^{-\phi} = i\bar{S}$, on which the $SL(2, \mathbb{Z})$ S-duality of type IIB acts by fractional linear transformations; formulas taken from there are converted with $\tau = i\bar{S}$. With an index, τ_i will denote the volumes of four-cycles, the real parts of the T_i. Gauge couplings are $\alpha = g^2/4\pi$; the hypercharge coupling is GUT-normalised, $\alpha_1 = \frac{5}{3}\alpha_Y$.

Supergravity. Four-dimensional $\mathcal{N} = 1$ supergravity, with Kähler potential K and superpotential W, is written in the conventions of Wess and Bagger.

Chapter 1

From ten to four dimensions: scales and supersymmetry

Lecture of 5 November 2026

This first lecture of the second part of the course is an overview. Its purpose is to give a global picture of how one goes from ten-dimensional superstring theory to the four-dimensional world we observe, before the following lectures fill in the details. Two questions organise it. The first is quantitative: when six of the ten dimensions are compact, which energy scales appear, and in which window of energies can we trust which effective description? The answer is a hierarchy of three scales, the Planck mass, the string scale and the Kaluza–Klein scale, controlled by two numbers, the string coupling g_s and the volume $\mathcal{V}$ of the extra dimensions in string units. The second question is qualitative: how much of the supersymmetry of ten-dimensional string theory survives in four dimensions? The answer depends on the holonomy of the compact space, and it is what singles out Calabi–Yau manifolds.

1.1 Where we stand and where we are going

The first part of the course, Prof. Ling Lin's module, described the string from the point of view of its worldsheet. The fundamental object is a vibrating string, and quantising it means quantising the two-dimensional field theory that lives on the surface it sweeps out, the worldsheet, whose fields are the spacetime coordinates X^M of the string. This is a two-dimensional conformal field theory. It has a Weyl symmetry, and the requirement that this symmetry has no anomaly fixes the dimension of spacetime, $D = 26$ for the bosonic string. The spectrum of the bosonic string contains a tachyon, a sign that the theory is unstable. To get rid of it one adds fermions on the worldsheet. The result is the superstring, whose critical dimension is $D = 10$. The Gliozzi–Scherk–Olive (GSO) projection removes the tachyon, and the theory turns out to be supersymmetric not only on the worldsheet but also in spacetime. The quantum states of the string form a tower of modes, a few of them massless and the others massive, and each of them corresponds to a spacetime field; among the massless modes of the closed string are the metric G_{MN}, the Kalb–Ramond two-form B_{MN} and the dilaton ϕ.

Besides closed strings there are open strings, whose endpoints are attached to extended objects, the Dp-branes, of any dimension up to that of spacetime. The open strings ending on a brane describe fields that live on it, in particular gauge fields, and on a stack of N coincident branes they give a non-abelian gauge theory with gauge group $U(N)$. There are five consistent perturbative superstring theories, which differ in how the spectrum is chosen:

type I, types IIA and IIB, in which worldsheet fermions are added for both the left- and the right-moving degrees of freedom of the closed string, and the two heterotic theories, with gauge groups $SO(32)$ and $E_8 \times E_8$, in which only the right-movers are supersymmetrised. Perturbatively they look different, but they are related by dualities, some of which exchange a strongly coupled theory with a weakly coupled one, and they are understood as different limits of a single underlying theory, M-theory (Section 2.2). Loop amplitudes are sums over Riemann surfaces of increasing genus, which replace the Feynman diagrams of field theory; since the string has a finite size, which acts as a natural cutoff, they have no ultraviolet divergences.

From here one could follow many directions: the quantum properties of each of the five theories, their mathematical structure, black holes in ten or fewer dimensions. This part of the course follows one of them, *string compactifications*: how to reproduce our four-dimensional world, with the Standard Model and cosmology, starting from ten-dimensional string theory. The idea is to take spacetime to be four-dimensional Minkowski space times a compact six-dimensional space, and to go to energies below the inverse size of the extra dimensions, where they can no longer be seen. What remains is a four-dimensional effective theory with local supersymmetry, a supergravity theory of the kind met in the supersymmetry course, with the important difference that it is not written down by hand: it is derived from a consistent theory of quantum gravity. Its physics depends strongly on the geometry and the topology of the extra dimensions, so we shall need some differential geometry and Calabi–Yau manifolds, and D-branes will play an essential role in reproducing the non-abelian gauge theories seen in nature. The module covers the low-energy effective action of string theory, D-brane actions and dualities, type II compactifications on Calabi–Yau manifolds, flux compactifications and moduli stabilisation, supersymmetry breaking in string models, and the basic principles of string cosmology.

We shall see that the way down from ten to four dimensions is not unique. There is a *landscape*, or a multiverse, of solutions, each corresponding to a different four-dimensional universe. The goal is to find at least one that reproduces what we know, the Standard Model and cosmology, and possibly also offers solutions to the open problems recalled at the beginning of the supersymmetry course: inflation, dark matter, dark energy, the hierarchy problem and supersymmetry breaking.

The subject is vast, and this part of the course is necessarily dense. Rather than deriving everything systematically from a set of assumptions, which would cover one topic in depth but give a very partial view of the subject, the aim is a global understanding of the whole picture, with details to be studied afterwards where one is interested. The first lectures are therefore largely qualitative, meant to build intuition for how things go; later lectures return to the main steps with explicit computations, together with the mathematical background they need (differential forms, complex manifolds, branes and the dualities that generalise electric–magnetic duality).

References. The following books, reviews and lecture notes cover the material of this part of the course, and much more:

1. L. E. Ibáñez and A. M. Uranga, *String Theory and Particle Physics: An Introduction to String Phenomenology* [1];

2. A. Hebecker, *Naturalness, String Landscape and Multiverse*, lecture notes freely available online [2];

3. R. Blumenhagen, B. Körs, D. Lüst and S. Stieberger, *Four-dimensional string compactifications with D-branes, orientifolds and fluxes*, a long review [3];

4. lecture notes by J. Louis, available on his website [4];

5. D. Baumann and L. McAllister, *Inflation and String Theory*, a book whose lecture-note version is available online [5];

6. M. Cicoli, J. P. Conlon, A. Maharana, S. Parameswaran, F. Quevedo and I. Zavala, *String cosmology: from the early universe to today* [6].

The first four are oriented towards particle physics and the low-energy limit of string theory. The last two are oriented towards cosmology, which is a particularly promising arena for testing string theory, because some cosmological questions, inflation above all, can only be addressed reliably in an ultraviolet-complete theory. The review [6] starts with an introduction to compactifications and then follows the whole cosmological history, from inflation through reheating to dark matter and dark energy.

The exam consists of a twelve-minute slide presentation on a topic chosen by the student, from either part of the course, followed by questions on that topic. Reproducing a topic as it was presented in the lectures can earn up to 29/30; for 30 or 30 cum laude something more is expected, for which references can be asked to the lecturers. A mark can be turned down once.

1.2 Superstring theory as a theory of quantum gravity

Superstring theory is a consistent quantum theory, and the reason for the interest in it is that it is a consistent theory of quantum *gravity*. Its massless spectrum contains a symmetric tensor field $G_{MN}(X)$, the metric: there is a massless spin-two particle, the graviton, and so there is gravity, in a theory whose loop amplitudes are finite. But string theory is not only a theory of gravity. It also contains gauge interactions and matter, basically all the rest of physics, so that it is a candidate for a unified description of all interactions, and it is a framework large enough that other approaches to quantum gravity may turn out to be contained in it.

The theory is consistent because two potential problems are absent. There are no tachyons, thanks to the GSO projection, and there are no states of negative norm. The second requirement is the absence of a Weyl anomaly on the worldsheet, and it fixes the number of spacetime dimensions to the critical dimension

$$D = 10. \tag{1.1}$$

String theory is the only theory we know that predicts the number of spacetime dimensions. The prediction is, at first sight, wrong, since we see four dimensions. But it is not so bad: the answer is a positive integer, and it could have been worse.

Where $D = 10$ comes from. In the covariant quantisation of the superstring the worldsheet theory contains, besides the D bosons X^M and their D fermionic partners, the ghosts that come from fixing the worldsheet diffeomorphisms and local supersymmetry. Each field contributes to the central charge of the Virasoro algebra: 1 for each boson, $\frac{1}{2}$ for each Majorana fermion, -26 for the bc ghosts and +11 for the $\beta\gamma$ superghosts. The Weyl anomaly is proportional to the total, $c_{\text{tot}} = \frac{3}{2}D - 15$, and vanishes only for $D = 10$, i.e. for a "matter" theory of central charge $c = 15$. The same condition makes the BRST charge nilpotent, $Q_B^2 = 0$, which is what guarantees that the negative-norm states introduced by covariant quantisation decouple from the physical spectrum. For the bosonic string the same counting gives $D - 26 = 0$.

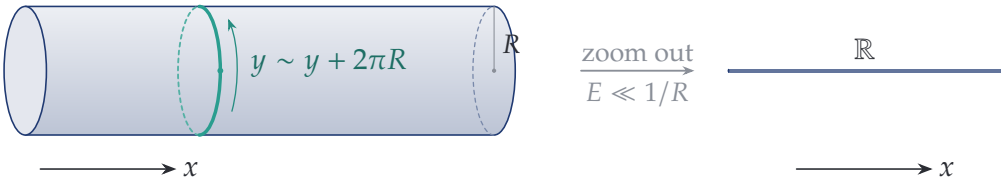


Figure 1.1: A cylinder $\mathbb{R} \times S^1$ seen from close by and from far away: at distances much larger than its circumference, or energies much lower than $1/R$, it looks like a line. Every point of the line carries a small circle, as every point of $\mathbb{R}^{1,3}$ carries a copy of Y_6.

1.3 Compactification and the Kaluza–Klein scale

To make contact with the real world we must study *string compactifications*. Ten-dimensional Minkowski space $\mathbb{R}^{1,9}$ is a solution of string theory, but there are others, in which ten-dimensional spacetime is the product of four-dimensional Minkowski space and a six-dimensional manifold,

$$X_{10} = \mathbb{R}^{1,3} \times Y_6, \tag{1.2}$$

where Y_6 is a *compact* space: a six-sphere, a six-dimensional torus, or something more complicated. Its volume $\mathrm{Vol}(Y_6)$ has dimensions of a length to the sixth power, so its $-1/6$ power is an energy, the *Kaluza–Klein scale*,

$$M_{KK} := \frac{1}{\mathrm{Vol}(Y_6)^{1/6}}. \tag{1.3}$$

M_{KK} is the energy above which one starts to see the extra dimensions; equivalently, $1/M_{KK}$ is the length scale below which they become visible. We expect

$$M_{KK} \gg E_{\mathrm{LHC}} \sim 1\,\mathrm{TeV},$$

the energies probed at the LHC, and this explains why we have not seen extra dimensions.

The idea is easiest to picture in two dimensions. Take a spacetime that is a cylinder, with one extended direction and one direction curled up into a circle, and look at it from very far away: one sees a line, not the circle (Figure 1.1). In the same way, it is not so strange to expect that at every point of our four-dimensional spacetime there is a six-dimensional curled-up space that we do not see, simply because it is too small.

The Kaluza–Klein tower. The name of M_{KK} comes from the tower of four-dimensional particles that a single higher-dimensional field gives, as met in the supersymmetry course. For a massless scalar $\Phi(x, y)$ on $\mathbb{R}^{1,3} \times S^1$, with $y \sim y + 2\pi R$, the Fourier expansion $\Phi = \sum_{n\in\mathbb{Z}} \varphi_n(x)\, e^{iny/R}$ turns the five-dimensional wave equation into four-dimensional Klein–Gordon equations for the modes φ_n with masses $m_n = |n|/R$. At energies $E \ll 1/R$ only the zero mode φ_0 can be produced, and the theory looks four-dimensional. On a six-dimensional torus with six circles of radius R, $\mathrm{Vol}(Y_6) = (2\pi R)^6$ and the lightest massive modes have $m = 1/R = 2\pi/\mathrm{Vol}(Y_6)^{1/6}$. Thus Equation (1.3) fixes M_{KK} only up to a factor of order one (here 2π), and for a less symmetric Y_6 the precise spectrum depends on its shape, not only on its volume; what matters for us is the scaling with the volume.

1.4 The string scale and the Planck scale

Let us now place the relevant scales on an energy axis. Somewhere high up is the four-dimensional Planck mass M_P; somewhere below it is the *string scale*, the inverse of the string

length,

$$M_s = \frac{1}{\ell_s}, \qquad \ell_s = 2\pi\sqrt{\alpha'}. \tag{1.4}$$

The string length is set by the only free parameter of string theory, α', which determines the string tension, $T = 1/(2\pi\alpha')$. When the string is quantised, its vibrational modes give states that are either massless or have masses set by $1/\sqrt{\alpha'} = 2\pi M_s$, growing with the level of excitation. Why have we drawn M_s below M_P? To answer this, we have to relate the two.

1.4.1 The ten-dimensional action and the dilaton

From the scattering amplitudes of strings, at tree level and at loop level, one can reconstruct the effective action of the massless modes of the string in ten dimensions. Among its terms is the ten-dimensional Einstein–Hilbert term

$$S_{10} \supset \frac{1}{2\kappa_{10}^2} \int \mathrm{d}^{10}X \, \sqrt{-G} \, e^{-2\phi} \, R_{10}, \qquad \frac{1}{2\kappa_{10}^2} = \frac{2\pi}{\ell_s^8} = 2\pi \, M_s^8, \tag{1.5}$$

where R_{10} is the Ricci scalar built out of the ten-dimensional metric G_{MN}, and ϕ is the *dilaton*, another massless mode of the string. The factor $e^{-2\phi}$ multiplying the Ricci scalar means that this is the action in the *string frame* (see the conventions).

The remarkable property of the dilaton is that its vacuum expectation value fixes the strength of the interactions among strings, the *string coupling*,

$$g_s = e^{\langle\phi\rangle}. \tag{1.6}$$

The string coupling is therefore not a parameter that one can choose by hand, as one chooses α'. It is a function of the expectation value of a field: when the dilaton moves, the strength of the string interactions changes.

The action in Equation (1.5) is the tree-level contribution. As usual, it receives perturbative loop corrections, and on top of those non-perturbative ones, and their size relative to the tree-level action is controlled by g_s. If the string coupling is large the loop corrections are large, and the tree-level action cannot be trusted. If instead $\langle\phi\rangle$ is large and negative, the string coupling is small, and in the limit $\langle\phi\rangle \to -\infty$ all loop corrections become negligible. If one did not want to assume anything, one would have to compute all possible quantum corrections, beyond tree level, perturbative and non-perturbative, and then see what dynamics the complete theory gives: this would really be a dream, and it is not doable in practice. What one does instead is to *assume* that the string coupling is small, so that to a first approximation the tree-level part is enough, and then check a posteriori that the potential for ϕ has a minimum where g_s is indeed small. Since the value of ϕ is determined dynamically, this is a consistency condition, not a choice; if it fails, loops should have been included from the very beginning. We shall see that there is a way to make sure that the condition holds, which is nice. In summary, to trust perturbation theory we need

$$g_s \ll 1 \qquad \Longleftrightarrow \qquad \langle\phi\rangle \to -\infty. \tag{1.7}$$

Why $e^{-2\phi}$, and the Einstein frame. In string perturbation theory a closed-string amplitude on a worldsheet of genus g (a sphere with g handles) is weighted by $g_s^{2g-2} = g_s^{-\chi}$, where $\chi = 2 - 2g$ is the Euler characteristic. The tree-level action comes from the sphere, $\chi = 2$, which is the origin of the factor $e^{-2\phi}$ in Equation (1.5); the one-loop terms (the torus, $\chi = 0$) come without dilaton factor, and so on. The kinetic term of G_{MN} is mixed with that of the dilaton through this factor. A Weyl rescaling $G_{MN} = e^{(\phi-\phi_0)/2} G^E_{MN}$, with $\phi_0 = \langle\phi\rangle$, removes it: in ten dimensions

continued

$\sqrt{-G}\,R_{10}[G] = e^{2(\phi-\phi_0)}\sqrt{-G_E}\,R_{10}[G_E]$ + (derivatives of ϕ), so that the factor $e^{-2\phi}$ becomes the constant $e^{-2\phi_0} = g_s^{-2}$ and the action takes the Einstein–Hilbert form in the Einstein-frame metric G^E_{MN}. With this choice the two frames coincide in the vacuum. For completeness, the full bosonic action of the NS–NS fields in the string frame is

$$S_{\rm NS} = \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}X\,\sqrt{-G}\,e^{-2\phi}\Big(R_{10} + 4\,\partial_M\phi\,\partial^M\phi - \frac{1}{2\cdot 3!}H_{MNP}H^{MNP}\Big),$$

with $H = \mathrm{d}B$ the field strength of the Kalb–Ramond field; the R–R fields and the fermions add further terms, which we shall meet in the following lectures.

1.4.2 Dimensional reduction and the Planck mass

We now perform the *dimensional reduction* of Equation (1.5) on the background of Equation (1.2), going to energies below the Kaluza–Klein scale. The ten-dimensional metric factorises into a four-dimensional piece and a six-dimensional one,

$$G_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = g_{\mu\nu}(x)\,\mathrm{d}x^\mu\mathrm{d}x^\nu + g_{mn}(y)\,\mathrm{d}y^m\mathrm{d}y^n, \tag{1.8}$$

so that $\sqrt{-G} = \sqrt{-g_4}\,\sqrt{g_6}$ and $R_{10} = R_4 + R_6$. Keeping the term with the four-dimensional Ricci scalar, the action becomes

$$S_{10} \supset 2\pi M_s^8 \underbrace{\int_{Y_6}\mathrm{d}^6y\,\sqrt{g_6}}_{\mathrm{Vol}(Y_6)} \int \mathrm{d}^4x\,\sqrt{-g_4}\,e^{-2\phi}\,R_4. \tag{1.9}$$

There are other contributions, in particular the one with the internal Ricci scalar R_6, which we ignore for now (see the addition below). In the vacuum the dilaton takes its expectation value and can be taken outside the integral:

$$S_{10} \supset \underbrace{2\pi M_s^8\,\mathrm{Vol}(Y_6)\,e^{-2\langle\phi\rangle}}_{=M_P^2/2}\int \mathrm{d}^4x\,\sqrt{-g_4}\,R_4. \tag{1.10}$$

This is the usual four-dimensional Einstein–Hilbert action, and its coefficient is the Planck mass, $M_P^2/2$. Writing the volume in terms of a dimensionless quantity $\mathcal{V}$, the volume in string units, $\mathrm{Vol}(Y_6) = \mathcal{V}\,\ell_s^6$, and using $e^{-2\langle\phi\rangle} = g_s^{-2}$ and $\ell_s^6 = M_s^{-6}$, we find

$$\boxed{M_P^2 = 4\pi\,\frac{\mathcal{V}}{g_s^2}\,M_s^2} \qquad\Longrightarrow\qquad M_s = \frac{g_s}{\sqrt{4\pi\mathcal{V}}}\,M_P \simeq g_s\,\frac{M_P}{\sqrt{\mathcal{V}}}. \tag{1.11}$$

Since ℓ_s is the only parameter of string theory, Equation (1.11) allows us to trade it for the Planck mass. Conceptually M_s is the fundamental scale and M_P a derived one, but M_P is the one we know, so in practice we fix $M_P \simeq 2.4\times10^{18}$ GeV and derive everything else from it. The other two quantities in Equation (1.11) are not parameters at all: g_s is the expectation value of the dilaton, and we shall see that $\mathcal{V}$ is the expectation value of another field, the volume modulus. In the end everything is derived dynamically.

What was dropped, and the Einstein frame in four dimensions. Three simplifications are hidden in Equation (1.9). First, the product form of the metric, Equation (1.8), ignores the possibility that the size of $\mathbb{R}^{1,3}$ depends on the point of Y_6 (a warp factor), which will become important in

continued

flux compactifications. Second, the term $\int_{Y_6} \mathrm{d}^6 y \sqrt{g_6}\, R_6$, which we dropped, contributes to the four-dimensional scalar potential; it vanishes when Y_6 is Ricci-flat, as Calabi–Yau manifolds are (Section 1.7). Third, we have set the dilaton and the volume to constants. If they are allowed to vary along $\mathbb{R}^{1,3}$, as the fields they are, the coefficient of R_4 in Equation (1.9) is $2\pi M_s^2\, \mathcal{V}(x)\, e^{-2\phi(x)}$, which is not constant, and the four-dimensional action is not in the Einstein frame. A Weyl rescaling of the four-dimensional metric by $e^{-2\phi}\mathcal{V}$, relative to its vacuum value, brings it to that frame; it is the origin of the powers of $\mathcal{V}$ in the four-dimensional scalar potential that we shall meet later.

The ten-dimensional Planck mass. In the ten-dimensional Einstein frame the coefficient of the Einstein–Hilbert term defines a ten-dimensional Planck mass, $\frac{1}{2\kappa_{10}^2 g_s^2} \equiv \frac{1}{2} M_{10}^8$, i.e.

$$M_{10}^8 = \frac{4\pi}{g_s^2}\, M_s^8, \qquad M_{10} = (4\pi)^{1/8}\, g_s^{-1/4}\, M_s, \qquad M_P^2 = M_{10}^8\, \mathrm{Vol}(Y_6). \tag{1.12}$$

The last relation is the general statement that the four-dimensional strength of gravity is diluted by the volume of the extra dimensions. Above M_{KK} gravity is ten-dimensional, and it is M_{10}, not M_P, that sets the scale at which it becomes strong. For $g_s < 1$ and $\mathcal{V} > 1$ it lies between the two: $M_s < M_{10} < M_P$.

1.5 Perturbative control: string loops and α' corrections

Equation (1.11) tells us that

$$M_s \ll M_P \simeq 2.4 \times 10^{18}\,\mathrm{GeV} \qquad \text{for} \qquad g_s \ll 1 \quad \text{and} \quad \mathcal{V} \gg 1. \tag{1.13}$$

(The board rounds M_P to $10^{18}\,\mathrm{GeV}$.) The first condition is that of Equation (1.7), needed to trust the string loop expansion. The second condition, that the volume be large in string units, is needed to trust another kind of perturbation theory. To see why, write the inverse volume as

$$\frac{1}{\mathcal{V}} = \frac{\ell_s^6}{\mathrm{Vol}(Y_6)} = \frac{(2\pi)^6\, \alpha'^3}{\mathrm{Vol}(Y_6)}. \tag{1.14}$$

The limit of large volume is the limit in which α' is small compared with the size of the extra dimensions, $\alpha'^3 \ll \mathrm{Vol}(Y_6)$. A string of fixed size moving in a volume that is huge compared with it is the same as a volume of fixed size in which the string length is sent to zero. But $\ell_s \to 0$ is the limit in which strings become point-like particles, and in which the standard Feynman diagrams of quantum field theory are recovered from the string diagrams. Corrections in $1/\mathcal{V}$ are therefore genuinely *stringy* corrections: they tell us that the fundamental objects are strings, not point-like particles.

So the effective theory receives two kinds of corrections, controlled by two different expansion parameters. The first are *string loops*, the sums over Riemann surfaces of higher genus, with expansion parameter g_s; in the point-particle limit they reduce to the loop diagrams of quantum field theory. The second, with expansion parameter $1/\mathcal{V}$, do not look like loop corrections from the point of view of the spacetime quantum field theory. They are tree-level corrections, but with higher derivatives, like the operators of dimension five, six, seven and higher that one generates at tree level in field theory by integrating out heavy particles; here the heavy modes that are integrated out are the massive string modes. They are called α' *corrections*, and controlling them requires $1/\mathcal{V} \ll 1$. From the point of view of spacetime they are higher-derivative corrections; from the point of view of the worldsheet theory they are *worldsheet loops*, the loops of the two-dimensional field theory, which are counted by α'. In the regime $g_s \ll 1, \mathcal{V} \gg 1$ one can trust the effective theory and neglect both string loops and higher-derivative corrections, at least to a first approximation.

The size of the α' corrections. Each power of α' in the ten-dimensional effective action comes with two more derivatives. On Y_6, of typical length $L = \mathrm{Vol}(Y_6)^{1/6}$, two derivatives cost $1/L^2$, so each order is suppressed by $\alpha'/L^2 = 1/\big((2\pi)^2\mathcal{V}^{1/3}\big)$. In type II string theory supersymmetry forbids the corrections at orders α' and α'^2, and the first correction to the gravitational action is a term of the form α'^3R^4, quartic in the curvature. Relative to the tree-level term it is of order $\alpha'^3/L^6 = 1/\big((2\pi)^6\mathcal{V}\big)$, which is precisely the quantity of Equation (1.14). Its effect on the four-dimensional theory, computed by Becker, Becker, Haack and Louis [10], will be an essential ingredient of the Large Volume Scenario. Strictly speaking, control of the α' expansion requires all the curvature radii of Y_6 and the sizes of all of its cycles, not only its total volume, to be large in string units.

1.6 A hierarchy of scales and of effective theories

We can now complete the picture of the energy scales, shown in Figure 1.2. At the top is $M_P \simeq 2.4 \times 10^{18}\,\mathrm{GeV}$, and the string scale is below it, at a value we do not know, since it depends on g_s and $\mathcal{V}$ through Equation (1.11). At energies above M_s we see the whole ten-dimensional string theory: all the string resonances can be produced. The theory we shall study is type IIB string theory, which at present looks the most promising for obtaining phenomenology from string theory, for reasons that start to be explained in Chapter 2.

At energies below the string scale one remains in ten dimensions, but one can no longer produce the massive string modes, only the massless ones. What is left is no longer a string theory but a quantum field theory of point particles in ten dimensions, the fields that correspond to the massless modes of the string, with spectrum and interactions fixed by string theory. Since we started from a superstring this theory is supersymmetric, and since it contains gravity its supersymmetry is local: it is a ten-dimensional supergravity. For the type IIB string it is *type IIB supergravity*, with ten-dimensional $\mathcal{N} = 2$: there are two supersymmetry generators, two Majorana–Weyl spinors of the same chirality. "Type IIB" means that the field content is the massless spectrum of the type IIB string, not something chosen by hand.

Going to still lower energies we reach the Kaluza–Klein scale, below which the extra dimensions are invisible and the theory is four-dimensional. By Equations (1.3) and (1.11),

$$M_{KK} = \frac{1}{\mathrm{Vol}(Y_6)^{1/6}} = \frac{M_s}{\mathcal{V}^{1/6}} \simeq g_s\,\frac{M_P}{\mathcal{V}^{2/3}}, \tag{1.15}$$

where the power of the volume is $\frac{1}{6} + \frac{1}{2} = \frac{2}{3}$, and, keeping the factor of Equation (1.11), $M_{KK} = g_sM_P/(\sqrt{4\pi}\,\mathcal{V}^{2/3})$. The Kaluza–Klein scale is below the string scale, $M_{KK} < M_s$: written in terms of M_P, the two scales are $g_sM_P/\mathcal{V}^{2/3}$ and $g_sM_P/\mathcal{V}^{1/2}$, and for large volume $\mathcal{V}^{2/3} > \mathcal{V}^{1/2}$; equivalently, $M_{KK}/M_s = \mathcal{V}^{-1/6} < 1$. In the regime of Equation (1.13), in which we trust the effective theory, the hierarchy of scales is therefore

$$M_{KK} < M_s < M_P, \qquad \frac{M_{KK}}{M_s} = \frac{1}{\mathcal{V}^{1/6}}, \qquad \frac{M_s}{M_P} = \frac{g_s}{\sqrt{4\pi\mathcal{V}}}. \tag{1.16}$$

Below M_{KK} one obtains a four-dimensional supergravity theory, described, as in global supersymmetry, by a Kähler potential and a superpotential.

As a reminder of the conventions: since our Einstein frame coincides with the string frame in the vacuum, $\mathcal{V}$ in Equations (1.11) and (1.15) is the Einstein-frame volume as well. In references that define the Einstein frame without subtracting ϕ_0 the same relations appear as $M_s = g_s^{1/4}M_P/\sqrt{4\pi\mathcal{V}'}$ and $M_{KK} \simeq M_P/\mathcal{V}'^{2/3}$, in terms of a volume $\mathcal{V}' = g_s^{-3/2}\mathcal{V}$ (see the conventions).

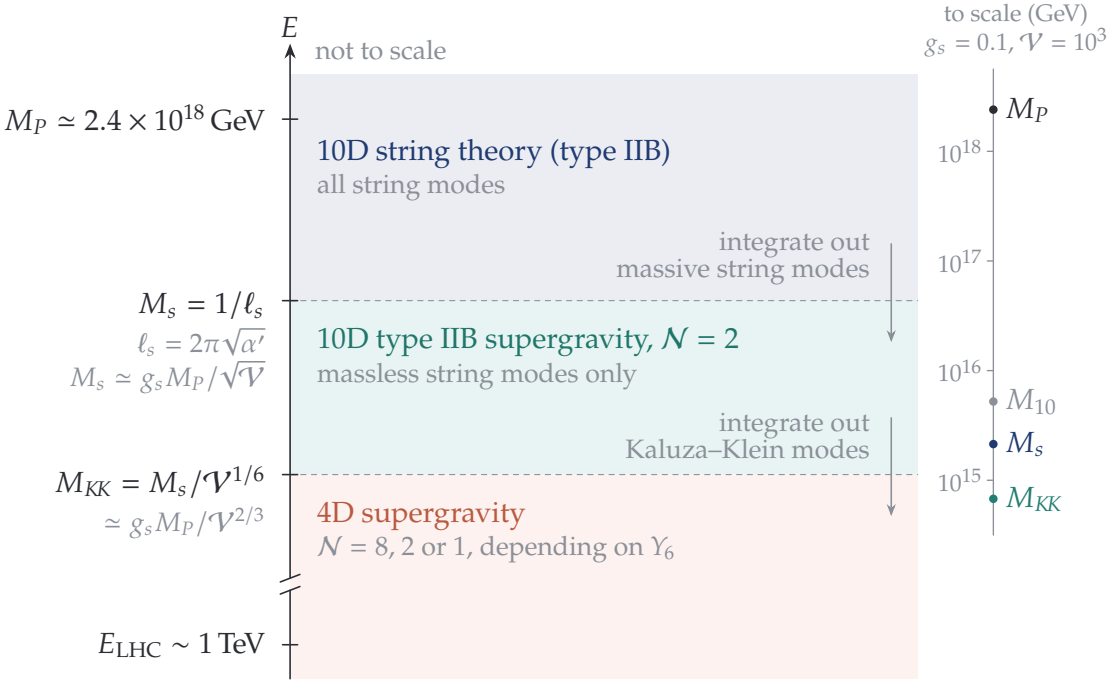


Figure 1.2: The hierarchy of scales of a string compactification and the effective description valid in each window of energies. The ratios are fixed by g_s and $\mathcal{V}$ through Equation (1.16).

1.7 How much supersymmetry survives: holonomy

The number of supersymmetries in four dimensions depends, of course, on the number of supersymmetries in ten dimensions, but it also depends on the topology of Y_6. The supersymmetry generators are spinors: in type IIB supergravity two ten-dimensional Majorana–Weyl spinors. Given one such spinor in ten dimensions, one can ask how many four-dimensional spinors it corresponds to, and the answer depends on the compact space. This is what singles out the famous Calabi–Yau manifolds. The statements in this section are made in words; we shall prove them later in this part of the course.

Consider first the simplest situation, a flat internal manifold, the six-dimensional torus T^6. In this case the largest amount of supersymmetry is preserved: one Majorana–Weyl spinor in ten dimensions corresponds to four spinors in four dimensions, and the two of type IIB give four-dimensional $\mathcal{N} = 8$ supergravity,

$$Y_6 = T^6, \quad \mathrm{Hol}(T^6) = \{\mathbb{1}\} \qquad \Longrightarrow \qquad \text{4D } \mathcal{N} = 8. \tag{1.17}$$

The property of T^6 that matters here is that it is a space of trivial *holonomy*: its holonomy group contains only the identity.

The holonomy is a simple notion. Take a manifold, choose a closed path in it starting and ending at a point p, take a vector at p and parallel-transport it around the path, as in Figure 1.3. When it gets back to p the vector has the same length, but it need not be the same vector. On a flat manifold it always comes back to itself, but on a curved manifold it is in general rotated. A different closed path gives in general a different rotation. The *holonomy group* is the set of all the transformations that relate the original vector to the transported one, for all possible closed paths. If the holonomy group contains only the identity, the vector always comes back to itself, as in the flat case; this is the trivial situation of the torus, which is flat.

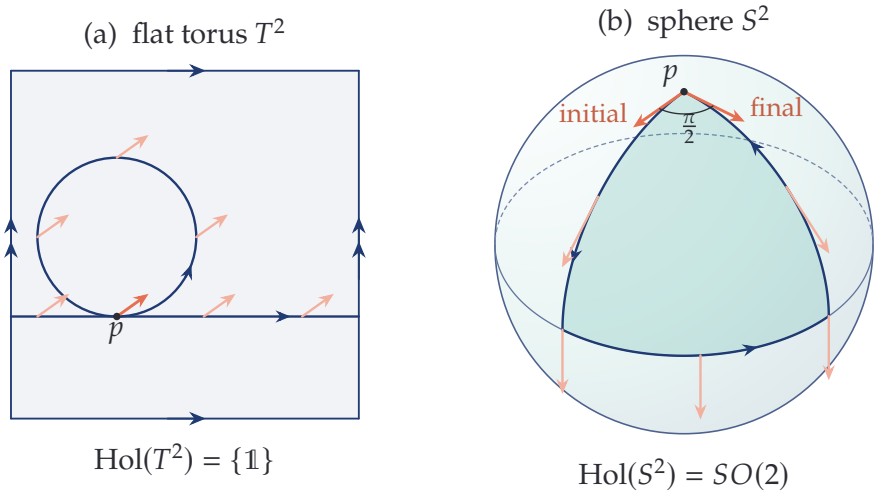


Figure 1.3: Parallel transport around closed loops. On a flat torus a vector always returns to itself, so the holonomy is trivial. On a sphere a vector carried around a closed loop returns rotated, here by $\pi/2$; the holonomy group collects all such rotations.

Now notice the beauty of this result. In the supersymmetry course we saw that $\mathcal{N} = 8$ is the largest amount of supersymmetry a four-dimensional theory can have: beyond it, the supermultiplets contain massless particles of spin higher than two, which have no consistent interactions and are not seen in nature. String theory never gives more than $\mathcal{N} = 8$, so a consistent theory reproduces this result automatically. Compactification of type IIB string theory on $\mathbb{R}^{1,3} \times T^6$ is a perfectly good solution of the theory. But it does not describe our world, because $\mathcal{N} = 8$ is too much supersymmetry. The world needs at most $\mathcal{N} = 1$ at high energies: in four-dimensional theories with $\mathcal{N} \geq 2$ the fermions cannot be chiral, while those of the Standard Model are (see also Section 1.8). The supersymmetry must be reduced, and the way to reduce it is what selects Calabi–Yau manifolds.

If instead of T^6 the space Y_6 is a *Calabi–Yau manifold*, whose holonomy group is $SU(3)$, then one Majorana–Weyl spinor in ten dimensions gives a single spinor in four dimensions, and type IIB gives four-dimensional $\mathcal{N} = 2$ supergravity:

$$Y_6 = \text{Calabi–Yau}, \quad \mathrm{Hol}(Y_6) = SU(3) \qquad \Longrightarrow \qquad \text{4D } \mathcal{N} = 2. \tag{1.18}$$

This is why Calabi–Yau manifolds, the six-dimensional manifolds of $SU(3)$ holonomy, became famous: they break the largest possible amount of supersymmetry while preserving some of it, which is what is needed to make contact with the real world.

Why the holonomy decides. On $\mathbb{R}^{1,3} \times Y_6$ the ten-dimensional Lorentz group is broken to $SO(1,3) \times SO(6)$, and the Majorana–Weyl representation **16** of $SO(1,9)$ decomposes as

$$\mathbf{16} = (\mathbf{2}, \mathbf{4}) \oplus (\bar{\mathbf{2}}, \bar{\mathbf{4}}), \tag{1.19}$$

where **2** and $\bar{\mathbf{2}}$ are the two Weyl spinors of $SO(1,3)$, and **4** and $\bar{\mathbf{4}}$ the two chiral spinors of $SO(6)$, the fundamental and antifundamental of $SU(4) \simeq \mathrm{Spin}(6)$. The Majorana condition relates the two terms, so a ten-dimensional supersymmetry parameter has the form $\epsilon(x, y) = \zeta(x) \otimes \eta(y) + \text{c.c.}$, with ζ a four-dimensional Weyl spinor and η a spinor on Y_6. A supersymmetry is preserved by the background if the variations of the fermions vanish in it. With no fluxes and a constant dilaton the only condition comes from the gravitino, $\delta\psi_M = \nabla_M \epsilon = 0$, which requires η to be covariantly constant on Y_6, $\nabla_m \eta = 0$. A covariantly constant spinor is unchanged by parallel transport around any loop, so it is a singlet of $\mathrm{Hol}(Y_6)$ acting on the **4**. Each such singlet η gives one four-dimensional supersymmetry for each ten-dimensional Majorana–Weyl generator.

continued

For T^6 all four components are singlets, and type IIB gives $2 \times 4 = 8$ supersymmetries. For Hol $= SU(3) \subset SU(4)$, $\mathbf{4} \to \mathbf{3} \oplus \mathbf{1}$ has a single singlet, and type II gives $\mathcal{N} = 2$; the heterotic and type I strings, which have a single Majorana–Weyl supersymmetry in ten dimensions, give $\mathcal{N} = 1$ directly, which is how Calabi–Yau manifolds first entered string theory [9] (Section 2.5). An intermediate case is Hol $= SU(2)$, e.g. $Y_6 = K3 \times T^2$, with $\mathbf{4} \to \mathbf{2} \oplus \mathbf{1} \oplus \mathbf{1}$ and $\mathcal{N} = 4$ from type II. Finally, a compact Kähler manifold of complex dimension three has Hol $\subseteq U(3)$, and Hol $\subseteq SU(3)$ is equivalent to Ricci-flatness; by Yau's theorem a Ricci-flat Kähler metric exists, in each Kähler class, precisely when the first Chern class vanishes. This is the usual definition of a Calabi–Yau manifold, and Ricci-flatness guarantees that the R_6 term dropped in Equation (1.9) vanishes.

To reach four-dimensional $\mathcal{N} = 1$ one more step is needed. Half of the degrees of freedom have to be projected away: one has to cut the spectrum. The operation that does it in type IIB is the *orientifold projection*. It reverses the orientation of the strings and identifies different modes of the string with each other, and keeping only the combinations invariant under it projects out half of the spectrum, and with it half of the supersymmetries. Type IIB string theory on a Calabi–Yau orientifold gives four-dimensional $\mathcal{N} = 1$ supergravity. We shall understand later in more detail what this operation means.

The orientifold projection. Type IIB string theory is invariant under worldsheet parity Ω_p, which reverses the orientation of the string and exchanges its left- and right-moving modes. Gauging this symmetry, i.e. keeping only the states invariant under it, gives type I string theory. In an orientifold, Ω_p is combined with an involution σ of the compact space, $\sigma^2 = 1$, and the theory is restricted to the states invariant under the combination; the name joins "orientation" and "orbifold". For a Calabi–Yau manifold σ is taken holomorphic and isometric, and two choices preserve four-dimensional $\mathcal{N} = 1$ [11]:

$$\mathcal{O} = (-1)^{F_L}\,\Omega_p\,\sigma^* \quad \text{with} \quad \sigma^*\Omega = -\Omega, \qquad\qquad \mathcal{O} = \Omega_p\,\sigma^* \quad \text{with} \quad \sigma^*\Omega = +\Omega,$$

where Ω is the holomorphic three-form of the Calabi–Yau and $(-1)^{F_L}$ the spacetime fermion number of the left-movers. The fixed loci of σ are *orientifold planes*: in the first case O3-planes, at points of Y_6, and O7-planes, wrapping four-cycles; in the second case O5- and O9-planes. Unlike D-branes, O-planes are not dynamical, and they carry negative tension and negative Ramond–Ramond charge. This will make them essential for the consistency (tadpole cancellation) of flux compactifications, and the O3/O7 case is the setting of the moduli-stabilisation scenarios studied later in the course.

All these statements hold at the compactification scale. This scale, as we have seen, is not known, but it is expected to be very high, perhaps 10^{18}, 10^{16} or 10^{15} GeV. At that scale, depending on the manifold on which the theory is compactified, one has four-dimensional $\mathcal{N} = 8$, $\mathcal{N} = 2$ for a Calabi–Yau, or $\mathcal{N} = 1$ for a Calabi–Yau orientifold. The remaining task is then to break the last $\mathcal{N} = 1$ to $\mathcal{N} = 0$, at low energy, around the TeV scale, and dynamically.

The plan of this part of the course is to explain these steps: start from ten-dimensional type IIB supergravity and see how to go down to four-dimensional theories with $\mathcal{N} = 8$, $\mathcal{N} = 2$ and finally $\mathcal{N} = 1$, discussing along the way the geometric notions, such as holonomy, that they require.

1.8 Questions and discussion

The lecture ended with questions from the students. We reproduce them here with the answers, because they touch on themes that will return throughout the course.

Question. *Do we have any reason to think that g_s is small?*

There is no a priori reason for the string coupling to be small. The region $g_s \ll 1$ is the

one in which we can perform computations, but nature does not need to be in the region where we can compute. There are, however, indications that it is small, and they come from observation: everything we see in nature so far looks weakly coupled. There are many small numbers in what we know. The electromagnetic coupling $\alpha_{\rm em} = e^2/4\pi$ is about $1/137$ at the energies of our experiments; the Yukawa couplings are small, that of the electron being $y_e \simeq 3 \times 10^{-6}$; in cosmology, the density fluctuations of the cosmic microwave background are of order 10^{-5}. We shall see that gauge and Yukawa couplings are functions of the string coupling (and of the other moduli), so their smallness is an indication that g_s is small.

But the dilaton is a field, and a field can have different configurations, different vacua. Some vacua may correspond to universes in which g_s is large. They would still be solutions of string theory, just not our universe, and in fact in most of the solutions we know g_s is not small. Once there are different solutions one must also understand, cosmologically, why one of them is realised in nature rather than another, or whether several are, in which case we are in a kind of multiverse and must ask how our universe is connected to the other parts. These are deep questions.

Small g_s is also required from the point of view of the theory, to trust the computations, so the phenomenological and the theoretical requirements point in the same direction. In principle one would start from ten-dimensional string theory and compute all quantum effects: a tree-level part, loop corrections and non-perturbative corrections. Dimensionally reducing the full result would then give the complete four-dimensional $\mathcal{N} = 1$ theory, with all quantum effects included. This is unfeasible. If the string coupling is small, however, one can keep only the leading-order result, knowing that the corrections one cannot compute are negligible. As said above, this is an assumption that has to be checked: the dilaton turns out to be a scalar field in four dimensions, one has to compute its potential and find where it has a minimum, since cosmologically the field is expected to end up there, and that minimum must lie at weak coupling. In most solutions this is not the case, but we shall see that in a subset of solutions it is guaranteed. We focus on those, both for the consistency of the theory and because they are good candidates to reproduce the Standard Model.

Question. *If we had started from type IIA, which is not chiral in ten dimensions, could we still obtain a chiral theory in four dimensions?*

Yes: one can also start from type IIA and obtain chiral four-dimensional theories. Chirality depends on the dimension: when we say that the world is chiral we mean that it is chiral in four dimensions, and a theory that is chiral in ten dimensions need not be chiral in four. Type IIB, for instance, is chiral in ten dimensions, but compactified on T^6 it gives four-dimensional $\mathcal{N} = 8$ supergravity, which is not chiral. A four-dimensional theory can be chiral only when it has $\mathcal{N} = 1$ (or no supersymmetry). Type IIA is not chiral in ten dimensions, but when it is compactified down to four-dimensional $\mathcal{N} = 1$ it can give a chiral theory as well. The reason why people mostly focus on type IIB rather than type IIA is that the latter is technically more complicated, not that it is excluded; we shall come back to this.

Question. *The real requirement of the worldsheet theory is not ten dimensions but a central charge $c = 15$. Are there options we are not considering, some conformal field theory that is not geometrical?*

Yes. The worldsheet conformal field theory describing the "matter" must have $c = 15$, and ten free bosons with their fermionic partners are only one way of achieving it. One should think of string theory, or M-theory, as a big space of theories in which one can move, finding different configurations. At some points the configuration can be described geometrically, as $\mathbb{R}^{1,9}$ or as $\mathbb{R}^{1,3} \times Y_6$; at other points there is no geometric description. An example is the limit in which the volume of the extra dimensions goes to zero, where one finds conformal field theory limits with no geometric interpretation.

Question. *Is there a reason to believe that our universe is not in one of these limits?*

Yes. In those limits there is no hierarchy. All scales are of order the string scale, and all couplings are of order one. To reproduce our universe one needs hierarchies: the string scale below the Planck scale, supersymmetry broken at a much lower scale, couplings that are small.

The counting behind $c = 15$ is the one of the addition after Equation (1.1): the ghosts contribute $-26 + 11 = -15$, so the rest of the worldsheet theory must contribute +15, with $\mathcal{N} = 1$ worldsheet supersymmetry. In a compactification $\mathbb{R}^{1,3} \times Y_6$ the four flat directions contribute $\frac{3}{2} \times 4 = 6$, and the internal theory must have $c = 9$. For a Calabi–Yau this is a supersymmetric sigma model on Y_6, but any conformal field theory with $c = 9$ (and, for spacetime supersymmetry, $\mathcal{N} = 2$ worldsheet supersymmetry) will do. A famous example is the Gepner model, built from $\mathcal{N} = 2$ minimal models, which describes the quintic Calabi–Yau at a point of small volume where the geometric description has broken down.

A remark of the lecturer's own belongs here. Everything said so far is put in by hand. We have assumed the background $\mathbb{R}^{1,3} \times Y_6$ and allowed deformations of Y_6: it can be a torus or a Calabi–Yau manifold, and we shall allow the Calabi–Yau to change in size, for example to have a different volume, or in shape. A change of shape at fixed size is like deforming a doughnut into a teacup: there is still one hole, one handle, and the volume may stay the same, but the shape changes. These deformations will be the moduli of the compactification (Section 2.7). It would be very nice to understand dynamically why we end up in this background rather than in something else, for example $\mathbb{R}^{1,4} \times Y_5$, but this is very hard, and we are not able to do it yet: it requires the whole non-perturbative M-theory, not perturbative string theory.

Question. *Why do we need to understand M-theory to determine Y_6?*

Think of the difference between first and second quantisation in quantum field theory. The first-quantised description is the worldline formalism, in which one follows one particle at a time; to describe as many particles as one likes one needs a quantum field theory. In the first part of the course the string was introduced as a generalisation of the worldline, the worldsheet, and this formulation of string theory is a kind of first-quantised version. What one would like is a string field theory, with a field whose excitations are strings, and a complete formulation of this kind, able to describe the theory non-perturbatively, is not available.

Question. *Would matrix theories, such as the BFSS model, count as string field theories?*

Matrix theories, such as the BFSS matrix model of Banks, Fischler, Shenker and Susskind, in which M-theory is described by the quantum mechanics of a large number of D0-branes, are proposals for the heart of M-theory, and branes play a crucial role in them; there are many papers on these proposals, but no consensus on the right formulation. The point is that what we can actually do is perturbative, and a simple way to see the problem is a quantum-mechanical system with two vacua. Everything we have done here studies one vacuum. Perturbation theory around it allows us to deform it, for instance to deform the Calabi–Yau, but never to reach the other vacuum, which could be, say, $\mathbb{R}^{1,4} \times Y_5$. What interpolates between two vacua in quantum mechanics is an instanton, a tunnelling effect, which is non-perturbative because it is not a small perturbation around either vacuum, and to describe it one needs the whole picture. Perturbative string theory around one vacuum gives, at low energy, one quantum field theory, for instance four-dimensional $\mathcal{N} = 1$ supergravity from type IIB; around another vacuum, another quantum field theory. Connecting the two, and understanding cosmologically how the different vacua were populated in the first place, would require a description that interpolates between the different solutions of string theory.

Question. *In such a description, should one see a smooth transition from, say, type IIB to the $E_8 \times E_8$ heterotic string?*

Yes, one should. But the situation is even more complicated than this. Even without leaving type IIB, string theory gives many, many solutions, the landscape. They were found within type IIB because this is the limit in which computations can be done and things are under control. Sticking to this setting, one finds many different quantum field theories, all based on Calabi–Yau manifolds (more precisely, orientifolds) in type IIB, differing by the fluxes that we shall meet later; and relating them to each other, even within type IIB, would require understanding all possible connections between vacua.

The present approach is more humble. Out of all the solutions, let us at least find one that looks like our universe: one with non-abelian gauge theories like the Standard Model, the right cosmology, and so on. There are many that look like our universe. The second question is what happens to the other solutions. This question can be asked regardless of string theory, and in quantum field theory the situation is even worse: one can write down infinitely many quantum field theories, changing the gauge group or the spectrum of fields, all perfectly viable as quantum field theories, but disconnected from one another; the choice is just ours. In string theory we do not yet know how many consistent solutions there are, but at the moment their number seems to be finite, and in principle one can understand, quantum mechanically and at the non-perturbative level, how they are connected. We are far from that. A more immediate programme, in the most promising setting for phenomenology, is to study all the possible solutions and understand their common features, and then to see whether they lead to predictions: if all of them contain a new particle that we should see in nature, for example, or if none of them does, one can confront this with data.

Chapter 2

String phenomenology: from the heterotic string to the moduli problem

Lecture of 6 November 2026

This lecture continues the general description, begun in Chapter 1, of how string theory can be connected with the real world. Like the first one it is mostly made of words; the equations will come in the following lectures. It has three parts. The first explains why we look for compactifications that preserve four-dimensional $\mathcal{N} = 1$ supersymmetry at the Kaluza–Klein scale. The second surveys the five superstring theories in ten dimensions, the dualities that relate them and their massless spectra, and uses them to explain why string phenomenology started with the heterotic string and why it later moved to type II theories. The third introduces a problem that every string compactification has to face, and around which much of this course is organised: the *moduli*, massless scalar fields that parametrise the size and the shape of the extra dimensions, and the need to stabilise them.

2.1 Why four-dimensional $\mathcal{N} = 1$ supersymmetry

Recall the chain of effective theories of Section 1.6 (Figure 1.2). We start from type IIB string theory, a theory of closed strings, and go to energies below the string scale, integrating out all the massive string modes. What is left is a ten-dimensional field theory, type IIB supergravity, whose field content is the massless spectrum of the string. We then go below the Kaluza–Klein scale. A massless field in ten dimensions looks, from the four-dimensional point of view, like a massless field plus a tower of Kaluza–Klein states (Section 1.3), and integrating out the tower we are left with four-dimensional fields only. The result is a four-dimensional supergravity theory, whose number of supersymmetries depends on the topology of the internal space Y_6 (Section 1.7). We pick Calabi–Yau manifolds, which we shall define properly from the mathematical point of view, because they break the largest amount of supersymmetry while preserving some of it, and, after the orientifold projection, lead to $\mathcal{N} = 1$ supersymmetry at the Kaluza–Klein scale.

One could also consider compactifications that break supersymmetry completely, so that the theory at the Kaluza–Klein scale has $\mathcal{N} = 0$, and try to do model building directly there. We do not. We focus on string compactifications with four-dimensional $\mathcal{N} = 1$ supersymmetry at the Kaluza–Klein scale, for four reasons.

The first is that $\mathcal{N} = 1$ is the only chiral version of four-dimensional supersymmetry: in theories with $\mathcal{N} \geq 2$ the fermions come in non-chiral representations (Section 1.7). With $\mathcal{N} = 1$ there is hope of obtaining the Standard Model, which is a chiral theory.

The second concerns the scale of supersymmetry breaking. We do not know the Kaluza–Klein scale. It could be at the edge of detection, close to the TeV scale, but more realistically it is very high, say 10^{14}–10^{15} GeV. The $\mathcal{N} = 1$ supersymmetry that survives there should then be broken spontaneously at much lower energies, for example at the TeV scale, to address the hierarchy problem. It should also be broken *dynamically*: the hope is that this happens through non-perturbative effects, as in confinement, which naturally generate scales exponentially smaller than the fundamental ones (see the addition at the end of Section 2.5). Supersymmetry broken at low energies has the good properties met in the supersymmetry course: it addresses the hierarchy problem, it provides a dark matter candidate, it leads to radiative electroweak symmetry breaking, in which the electroweak symmetry is broken dynamically by loop corrections, and it makes the gauge couplings unify. The hope is to recover all of these from a top-down approach. Whether this works is a property of the effective theory: one has to derive it from string theory, compute the non-perturbative effects, and check whether they indeed break the remaining $\mathcal{N} = 1$ at low energies. It might also turn out that this is never possible.

The third concerns how the breaking is communicated. Besides knowing whether supersymmetry is broken at low energies, one must understand how the breaking is mediated to the *visible sector*, the supersymmetric version of the Standard Model, for instance the minimal supersymmetric Standard Model (MSSM). Some other sector of the compactification is responsible for breaking supersymmetry, and it must couple to the MSSM in such a way that the MSSM feels the breaking and the supersymmetric partners become massive. The breaking felt by the MSSM must be *soft*, so that the cancellation of quadratic divergences survives, and soft supersymmetry breaking is generated naturally in supergravity theories derived from string theory, through gravitational interactions. This is *gravity mediation*.

These three reasons are phenomenological. The fourth is theoretical, and we have not met it yet in the supersymmetry course: $\mathcal{N} = 1$ *supersymmetry allows one to control the quantum corrections to the effective theory.* A supersymmetric theory is specified by its Kähler potential K and its superpotential W, and one can ask how these quantities behave under quantum corrections. Each of them has a tree-level part, loop corrections and non-perturbative corrections, but not on the same footing:

$$K = K_{\text{tree}} + K_{\text{loops}} + K_{\text{np}}, \qquad W = W_{\text{tree}} + W_{\text{np}}. \tag{2.1}$$

The superpotential is exact at tree level in perturbation theory: it receives no loop corrections, as a consequence of supersymmetry and of the fact that it is a holomorphic quantity; the only corrections beyond tree level are non-perturbative. This will be proved in the supersymmetry course. It is a very powerful result: if one computes W at tree level, one knows for sure that there are no loop corrections. The Kähler potential is different and does receive loop corrections. Since computing quantum corrections in string theory is extremely hard, this control is very important. In a compactification that breaks supersymmetry completely there is no such control, and it is much harder to compute the effective theory. Keeping some supersymmetry helps to compute the effective theory, but not too much of it: with $\mathcal{N} = 1$ and not $\mathcal{N} = 2$ the theory remains chiral and is still good for phenomenology. That is the balance.

Why W is not renormalised. The argument, due in this form to Seiberg, combines holomorphy with symmetries. Every coupling in W can be promoted to the expectation value of a background chiral superfield, and W must be a holomorphic function of the chiral superfields and of these background fields. In string compactifications the couplings are themselves the expectation values of moduli, and the imaginary parts of many moduli are axions: if $T = \tau + ib$, the field b descends from a p-form gauge field (for instance from C_4 integrated over a four-cycle), and gauge invariance of the p-form gives it a shift symmetry $b \to b + c$ that holds to all orders in perturbation theory. A holomorphic W can depend on τ only through T, hence also on b, and

continued

the shift symmetry forbids this: perturbatively W cannot depend on T at all. Non-perturbative effects (instantons, gaugino condensates) break the continuous shift symmetry to a discrete one and allow terms such as $W_{\text{np}} \propto e^{-aT}$, which are exponentially small at weak coupling or large volume. The Kähler potential is real and depends on $T + \bar{T}$, which the shift symmetry leaves invariant, so nothing protects it: it receives both string-loop and α' corrections, such as the α'^3 correction mentioned in Section 1.5. The holomorphic gauge kinetic function, which sets the gauge couplings, is protected in the same way as W, except for a one-loop correction. The interplay between a protected W, generated non-perturbatively, and a corrected K is the basis of the moduli-stabilisation scenarios studied later in the course.

This control is absent in $\mathcal{N} = 0$ compactifications, those that really break supersymmetry completely: one Majorana–Weyl spinor in ten dimensions then corresponds to no spinor at all in four dimensions, so a ten-dimensional supersymmetry generator gives nothing. An example is $Y_6 = S^6$, a six-dimensional sphere, whose holonomy group is $SU(4)$ in the lecture's words, to be compared with $SU(3)$ for a Calabi–Yau manifold. Why the holonomy group determines the number of four-dimensional supersymmetries was sketched in the addition at the end of Section 1.7, and will be explained towards the end of the course.

Question. *A Calabi–Yau fourfold, used in F-theory to compactify from twelve dimensions to four, also has $SU(4)$ holonomy. Does that also give $\mathcal{N} = 0$?*

The Calabi–Yau fourfold does have $SU(4)$ holonomy; in general a Calabi–Yau n-fold has $SU(n)$ holonomy. But in that case the twelve-dimensional compactification is a fictitious construction: there is no twelve-dimensional theory. What one actually does is to start from eleven dimensions and compactify on the fourfold, which has eight real dimensions, down to $11 - 8 = 3$ dimensions. The fourfolds used for this have a fibre structure, with a two-torus as the fibre; roughly speaking, one can see them as a Calabi–Yau threefold with a two-torus attached. To recover four dimensions one sends the volume of the torus to zero. The torus has two circles, and in the limit one of the directions does not disappear but grows to infinite size, through the dualities of Section 2.2, and becomes the fourth direction of Minkowski space. What remains is a six-dimensional space of the Calabi–Yau type times four-dimensional Minkowski space, so that in the end one is again compactifying on a space of $SU(3)$ holonomy, a Calabi–Yau threefold. This is very advanced material, which we shall not need.

$SU(4)$ holonomy in six and in eight dimensions. The common name $SU(4)$ hides two different situations, and the answer to the question is no: a Calabi–Yau fourfold preserves supersymmetry. On a generic oriented six-manifold, S^6 included, the holonomy is the whole $SO(6)$, and on spinors it acts as the whole of $\text{Spin}(6) \simeq SU(4)$, under which the chiral spinor is the fundamental $\mathbf{4}$ (Equation (1.19)). The $\mathbf{4}$ contains no singlet, so there is no covariantly constant spinor and no supersymmetry survives. (The round S^6 is not Ricci-flat, so on its own it does not even solve the ten-dimensional vacuum equations; it serves only as an example of a space without invariant spinors.) For a Calabi–Yau fourfold, instead, $SU(4)$ is a proper subgroup of the rotation group $SO(8)$ of the eight-dimensional tangent space, on which it acts as $\mathbf{4} \oplus \bar{\mathbf{4}}$. The two chiral spinors of $SO(8)$ decompose as $\mathbf{8}_s \to \mathbf{1} \oplus \mathbf{1} \oplus \mathbf{6}$ and $\mathbf{8}_c \to \mathbf{4} \oplus \bar{\mathbf{4}}$, so there are two invariant spinors. The 32 supercharges of M-theory, $\mathbf{32} \to (\mathbf{2}, \mathbf{8}_s) \oplus (\mathbf{2}, \mathbf{8}_c)$ under $SO(1,2) \times SO(8)$, leave $2 \times 2 = 4$ real supercharges: three-dimensional $\mathcal{N} = 2$, which is four-dimensional $\mathcal{N} = 1$ reduced on a circle.
The way back to four dimensions uses a duality: M-theory on a two-torus of area A is equivalent to type IIB on a circle whose radius grows as A shrinks, $R_{\text{IIB}} \propto 1/A$, with the type IIB axio-dilaton given by the complex structure of the torus (Section 2.7). Fibring the torus over a base B_3 and taking $A \to 0$ gives type IIB on $\mathbb{R}^{1,3} \times B_3$, with an axio-dilaton that varies over B_3. This is F-theory, whose "twelve dimensions" are the ten of type IIB plus the two of the auxiliary torus.

2.2 Five superstring theories and their dualities

Our starting point will be ten-dimensional type IIB string theory, because, as we shall try to explain, it is at present the most promising one for phenomenology. In the end the starting point does not matter much, since all string theories are related by dualities. What changes from one description to another is which quantities are easy to compute: something that looks weakly coupled, and so computable, in one description can look strongly coupled, and hard to compute, in another.

At the moment there are five consistent perturbative superstring theories in ten dimensions, met in the first part of the course: type IIB, type IIA, type I, and the two heterotic theories, with gauge groups $SO(32)$ and $E_8 \times E_8$. They are all related by dualities of two kinds. *S-dualities* are strong–weak coupling dualities: they act on the dilaton, and exchange a theory at string coupling g_s with one at coupling $1/g_s$. *T-dualities* act instead on the geometry: they change the radius of the compactification manifold. The five theories are therefore best thought of not as different theories but as different limits, or better different perturbative corners, of a single, more fundamental theory, *M-theory*. What we know of M-theory is its low-energy limit, eleven-dimensional supergravity. In it, the size of the eleventh dimension corresponds to the string coupling. In a perturbative corner, where the coupling goes to zero and perturbation theory can be used, the size of the eleventh dimension goes to zero, and one recovers a ten-dimensional theory; when the coupling becomes large, one starts to see the eleventh dimension.

The simplest example is the relation between M-theory and type IIA. Compactified on a circle S^1 of radius R, going from eleven to ten dimensions, M-theory gives type IIA string theory, and R is set by the dilaton in such a way that

$$R \to 0 \qquad \Longleftrightarrow \qquad g_s \to 0 \qquad \text{(type IIA limit)}. \tag{2.2}$$

Type IIA is in turn T-dual to type IIB. T-duality changes the size of the compactification manifold: type IIA compactified on a circle of radius R is the same theory as type IIB compactified on a circle of radius

$$\tilde{R} = \frac{\alpha'}{R} = \frac{1}{4\pi^2 M_s^2 R}. \tag{2.3}$$

The two are the same theory, with R exchanged for α'/R: an example of a duality. The whole web of dualities is shown in Figure 2.1.

The eleventh dimension, T-duality and the web. The radius of the M-theory circle is related to the type IIA coupling by

$$R_{11} = g_s \sqrt{\alpha'}.$$

One way to see it is Witten's. Type IIA string theory contains D0-branes, point particles of mass $1/(g_s\sqrt{\alpha'})$, and bound states of n of them, of mass $n/(g_s\sqrt{\alpha'})$. At strong coupling these become light, and they are precisely the tower of Kaluza–Klein momentum states n/R_{11} of a circle of radius $R_{11} = g_s\sqrt{\alpha'}$: at strong coupling an eleventh dimension opens up, and the low-energy theory is eleven-dimensional supergravity. Its Planck length is $\ell_{11} \propto g_s^{1/3}\sqrt{\alpha'}$, so that $R_{11}/\ell_{11} \propto g_s^{2/3}$, up to numerical factors that depend on conventions: weak coupling means a circle that is small in eleven-dimensional Planck units.

T-duality is a statement about the spectrum. A closed string on a circle of radius R has quantised momentum n/R and can wind w times around the circle, which costs an energy wR/α', i.e. the tension $1/(2\pi\alpha')$ times the length $2\pi wR$; the masses in the remaining dimensions are $M^2 = (n/R)^2 + (wR/\alpha')^2 +$ (oscillator contributions), with the level-matching condition relating nw to the oscillators. This spectrum, and in fact the whole interacting theory, is invariant under

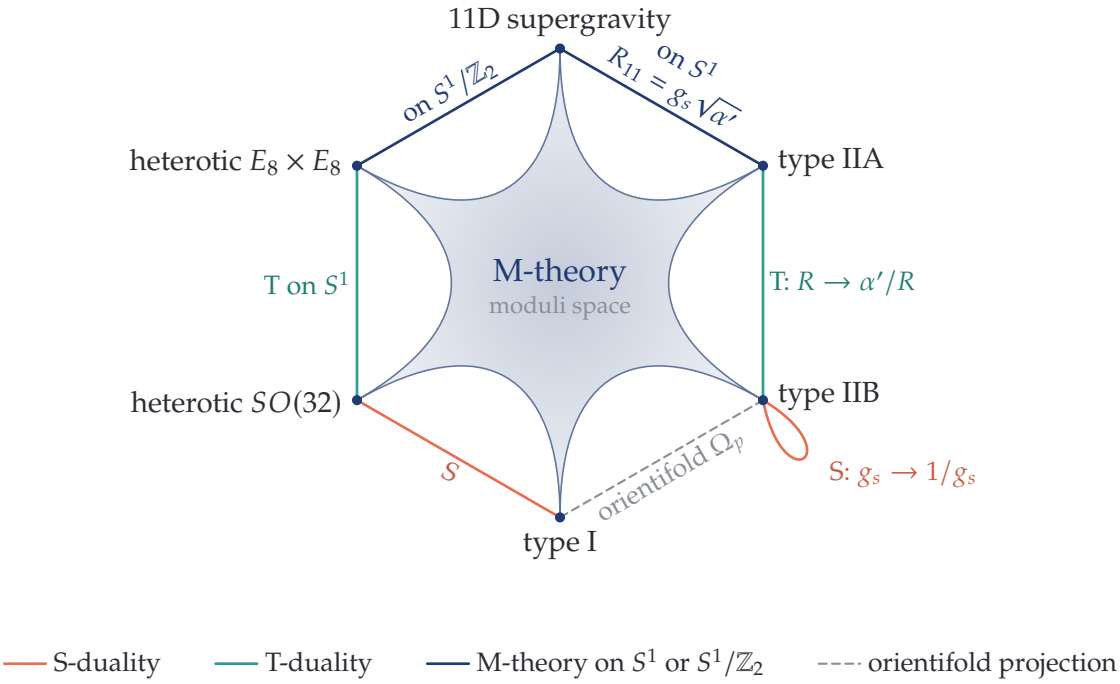


Figure 2.1: The five ten-dimensional superstring theories and eleven-dimensional supergravity as limits of a single theory, M-theory, and the dualities that relate them.

continued

$R \to \alpha'/R$ with $n \leftrightarrow w$, provided the string coupling also changes, $g_s \to g_s\sqrt{\alpha'}/R$, so that the lower-dimensional coupling is the same. In the superstring the transformation also reverses the chirality of the Ramond ground state of one side of the string, which turns type IIA into type IIB and vice versa. The self-dual radius is $R = \sqrt{\alpha'} = \ell_s/2\pi$.
The rest of the web of Figure 2.1 is as follows. Type IIB is self-dual under S-duality, which acts on the combination $C_0 + ie^{-\phi}$ of the dilaton and the R–R scalar (Sections 2.3 and 2.7) by $SL(2,\mathbb{Z})$ transformations, and maps $g_s \to 1/g_s$ when $C_0 = 0$. Type I, which is type IIB with the orientifold projection by worldsheet parity Ω_p (Section 1.7), is S-dual to the $SO(32)$ heterotic string. The two heterotic theories are T-dual to each other on a circle, with a Wilson line for the gauge field. Finally, the $E_8 \times E_8$ heterotic string at strong coupling is M-theory on an interval $S^1/\mathbb{Z}_2$, with one E_8 factor on each of its two boundaries (Hořava and Witten).

2.3 Massless spectra and the first string models

Historically, the first attempts at string phenomenology, the research area that tries to obtain our real world from string theory, started around 1985, in the mid-1980s. String theory was born at the end of the 1960s and developed in the 1970s, and in the mid-1980s people started trying to connect it to the real world. Among the pioneers was Fernando Quevedo, the lecturer's PhD supervisor, who started his PhD in the mid-1980s with Steven Weinberg, the Weinberg who received the Nobel Prize for the electroweak theory at the heart of the Standard Model. When Quevedo asked him for a research project, Weinberg suggested trying to obtain the Standard Model from string theory, a sign of Weinberg's interest in these questions, even though the two never wrote a paper together. Many others started working on this in the same years.

These initial attempts focused not on type IIB but on the $E_8 \times E_8$ heterotic string, because type II theories did not seem to give rise to non-abelian gauge bosons in four dimensions.

Sector	Type IIA	Type IIB	Heterotic
NS–NS	G_{MN}, B_{MN}, ϕ	G_{MN}, B_{MN}, ϕ	G_{MN}, B_{MN}, ϕ, A^a_M
R–R	C_1, C_3	C_0, C_2, C_4	—
NS–R	ψ^1_{M+}, λ^1_-	ψ^1_{M+}, λ^1_-	ψ_{M+}, λ_-, λ^a_+
R–NS	ψ^2_{M-}, λ^2_+	ψ^2_{M+}, λ^2_-	—
10D supersymmetry	$\mathcal{N} = 2$, non-chiral	$\mathcal{N} = 2$, chiral	$\mathcal{N} = 1$, chiral

Table 2.1: Massless spectra of the type IIA, type IIB and heterotic superstrings in ten dimensions, sorted by sector: the first label refers to the left-movers, the second to the right-movers (for the heterotic string see the text). G_{MN} is the metric, B_{MN} the Kalb–Ramond two-form, ϕ the dilaton, C_p the R–R p-forms, ψ_M the gravitini, λ the dilatini; A^a_M and λ^a are the heterotic gauge bosons and gaugini, $a = 1, \ldots, \dim G = 496$, with $G = SO(32)$ or $E_8 \times E_8$. The subscripts ± give the ten-dimensional chirality.

Looking at the spectrum of the type II theories, people did not see any non-abelian gauge bosons, which are needed for the Standard Model, and concluded that these theories, although consistent, were not good for the real world. The heterotic string, instead, has a gauge group in its massless spectrum, $E_8 \times E_8$, and E_8 is a huge group which contains the Standard Model gauge group as a subgroup. The idea was to break one E_8 factor down to the Standard Model with a Higgs mechanism, while the dynamics of the second E_8 factor could be used to break supersymmetry. This programme turned out to be very hard, for reasons that will be stressed below, although people still work on heterotic phenomenology today. Then, around 1995, came the discovery of *D-branes* as dynamical objects of string theory, and with it the realisation that the perturbations around a background containing D-branes, the open strings ending on them, can behave as non-abelian gauge bosons. D-branes are present in type II theories, so one had simply missed an ingredient: the non-abelian gauge bosons are absent from the closed-string spectrum, but they can come from the open-string spectrum. From 1995 onwards type II theories have been studied more than the heterotic string for phenomenology, even if there is still interest in heterotic constructions.

To see why, let us summarise the massless spectra: we focus on the massless modes of the string and ask which ten-dimensional fields they correspond to in spacetime. They are listed in Table 2.1 for type IIA, type IIB and the heterotic string.

The labels NS and R stand for Neveu–Schwarz and Ramond, and they are boundary conditions for the fermions on the worldsheet. The bosonic coordinates $X^M(\sigma)$ of a closed string must come back to themselves when one goes around the string, $\sigma \to \sigma + 2\pi$. When the theory is supersymmetrised one must add worldsheet fermions ψ^M, and these can come back either to themselves or to minus themselves:

$$\psi^M(\sigma + 2\pi) = +\psi^M(\sigma) \quad \text{(Ramond)}, \qquad \psi^M(\sigma + 2\pi) = -\psi^M(\sigma) \quad \text{(Neveu–Schwarz)}. \tag{2.4}$$

On a closed string there are left-movers and right-movers, and the condition can be chosen independently for each. In type II theories there are worldsheet fermions for both, and so two spacetime supersymmetries, one coming from the left-movers and one from the right-movers; this is why they are called type II, and why all four combinations NS–NS, R–R, NS–R and R–NS appear. In the heterotic string there are worldsheet fermions only for the right-movers, and so only one spacetime supersymmetry. The left-movers have no worldsheet fermions at all: they are those of the bosonic string. Being bosonic, they produce only spacetime bosons,

as an NS sector does, which is why the two heterotic sectors are labelled NS–NS and NS–R, according to the boundary condition of the right-movers.

Now the fields. In the bosonic sector there are always the metric G_{MN}, the antisymmetric Kalb–Ramond two-form B_{MN} and the dilaton ϕ, in the NS–NS sector: this is the spectrum of any bosonic string theory. The difference in the heterotic string is that there are also gauge bosons A^a_M, ten-dimensional vectors (hence the index M, not μ), and many of them: $a = 1, \dots, \dim G$, where the gauge group is $G = SO(32)$ or $G = E_8 \times E_8$. These groups are huge: $\dim E_8 = 248$, so that both have dimension 496. The E stands for exceptional: E_6, E_7 and E_8 are the exceptional Lie groups, related to the octonions, the generalisation of the complex numbers and the quaternions. This is why people expected the heterotic string to work: with non-abelian gauge bosons in ten dimensions, the four-dimensional theory would inherit some of them.

In the R–R sector type IIA has a one-form C_1 and a three-form C_3, and type IIB a zero-form C_0, a two-form C_2 and a four-form C_4. These are differential forms, i.e. antisymmetric tensors, such as $C_2 \leftrightarrow C_{MN} = C_{[MN]}$ and $C_4 \leftrightarrow C_{MNPQ} = C_{[MNPQ]}$, and C_0 is a pseudoscalar. They are all bosons. There are also higher forms, but they are dual to the ones listed: a p-form C_p in ten dimensions is equivalent to an $(8-p)$-form, since the field strengths are related by $F_{p+1} = \star F_{9-p}$, and in type IIB the field strength of C_4 is self-dual, $F_5 = \star F_5$.

The NS–R and R–NS sectors give fermions. There are two *gravitini* ψ_M, of spin 3/2: as we saw in the supersymmetry course, there are as many gravitini as supersymmetries, and type II theories have ten-dimensional $\mathcal{N} = 2$. The subscripts ± give the ten-dimensional chirality: a spinor has definite chirality if it is an eigenstate of $\Gamma_{11} = \Gamma^0\Gamma^1\cdots\Gamma^9$, the ten-dimensional analogue of γ_5, with eigenvalue ±1. There are also two *dilatini* λ, of spin 1/2, the supersymmetric partners of the dilaton. In type IIA the two gravitini have opposite chiralities, and so do the two dilatini: type IIA is a non-chiral theory in ten dimensions. In type IIB the two gravitini have the same chirality, and so do the two dilatini: type IIB is a chiral theory in ten dimensions (see also the conventions). The heterotic string has a single supersymmetry in ten dimensions, hence one gravitino and one dilatino, and in addition the supersymmetric partners of the gauge bosons, the *gaugini* λ^a: it is ten-dimensional $\mathcal{N} = 1$ supergravity coupled to super-Yang–Mills theory with gauge group G.

Where the table comes from. In light-cone gauge the massless states of each side of the closed superstring, after the GSO projection, form a representation of the little group $SO(8)$. In the NS sector the massless state is $\psi^i_{-1/2}|0\rangle$, the vector $\mathbf{8}_v$. In the R sector the periodic fermions have zero modes ψ^M_0 obeying a Clifford algebra, $\{\psi^M_0, \psi^N_0\} = \eta^{MN}$, so the ground state is a spacetime spinor, $\mathbf{8}_s$ or $\mathbf{8}_c$ depending on the GSO projection. This is why R–R states, products of two spinors, are bosons described by antisymmetric tensors, and why NS–R and R–NS states are fermions. The products are

NS–NS:	$\mathbf{8}_v \otimes \mathbf{8}_v = \mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_v$	$(\phi,\ B_{MN},\ G_{MN})$,
R–R, IIA:	$\mathbf{8}_s \otimes \mathbf{8}_c = \mathbf{8}_v \oplus \mathbf{56}_v$	$(C_1,\ C_3)$,
R–R, IIB:	$\mathbf{8}_s \otimes \mathbf{8}_s = \mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_+$	$(C_0,\ C_2,\ C_4$ with $F_5 = \star F_5)$,
NS–R:	$\mathbf{8}_v \otimes \mathbf{8}_s = \mathbf{8}_c \oplus \mathbf{56}_s$	$(\lambda^1,\ \psi^1_M)$,

and the R–NS sector gives $\mathbf{8}_s \otimes \mathbf{8}_v = \mathbf{8}_c \oplus \mathbf{56}_s$ in type IIB and $\mathbf{8}_c \otimes \mathbf{8}_v = \mathbf{8}_s \oplus \mathbf{56}_c$ in type IIA. Each gravitino comes with a dilatino of the opposite chirality; the two gravitini have the same chirality in type IIB and opposite chiralities in type IIA. Each sector has 64 states, so there are 128 bosons and 128 fermions, the type II supergravity multiplet.

In the heterotic string the right-movers give $\mathbf{8}_v \oplus \mathbf{8}_s$ as above. The left-movers are those of the bosonic string, in 26 dimensions: 10 of spacetime and 16 compactified on a torus $\mathbb{R}^{16}/\Gamma_{16}$. Consistency (modular invariance) requires Γ_{16} to be an even self-dual lattice, and in sixteen dimensions there are exactly two, the root lattice of $E_8 \times E_8$ and the lattice of $\mathrm{Spin}(32)/\mathbb{Z}_2$. The

continued

massless left-moving states are $\alpha^i_{-1}|0\rangle$, an $\mathbf{8}_v$, and 496 singlets of $SO(8)$: the 16 states $\alpha^I_{-1}|0\rangle$ of the internal bosons and the 480 states $|p\rangle$ with $p \in \Gamma_{16}$, $p^2 = 2$. Together they form the adjoint representation of G, of dimension $16 + 480 = 496$. The product $(\mathbf{8}_v \oplus \mathbf{8}_s) \otimes \mathbf{8}_v$ gives G_{MN}, B_{MN}, ϕ, the gravitino in $\mathbf{56}_s$ and the dilatino in $\mathbf{8}_c$; the product $(\mathbf{8}_v \oplus \mathbf{8}_s) \otimes \mathbf{496}$ gives the gauge bosons A^a_M and the gaugini in $\mathbf{8}_s$. The gaugini therefore have the chirality of the gravitino, opposite to that of the dilatino. Equivalently, in the supersymmetry transformations $\delta\phi \propto \bar\epsilon\lambda$ and $\delta A^a_M \propto \bar\epsilon\,\Gamma_M\lambda^a$ a Majorana–Weyl bilinear $\bar\epsilon\chi$ is non-zero only for opposite chiralities and $\bar\epsilon\,\Gamma_M\chi$ only for equal ones, so the dilatino has the chirality opposite to ϵ, and the gaugini, like the gravitino, the same.

2.4 Gauge bosons in four dimensions

The ten-dimensional gauge bosons of the heterotic string lead, in four dimensions, to four-dimensional gauge bosons: choosing the index $M = \mu$ and keeping the zero mode, constant along Y_6, one obtains $A^a_\mu(x)$, non-abelian gauge bosons that can give the gluons and the W and Z bosons of the Standard Model. In type IIA and type IIB this is not possible, and the reason is worth understanding, because it will come back in the discussion of moduli.

A ten-dimensional two-form, an object with two indices, could give objects with one index in four dimensions. Taking for instance $M = \mu$ along $\mathbb{R}^{1,3}$ and $N = n$ along Y_6, the component $B_{\mu n}$ is, from the four-dimensional point of view, a vector labelled by n. To extract its massless modes one expands it in the zero modes of the internal Laplacian, which for an object with one internal index are the harmonic one-forms $\omega^I_n(y)$ of Y_6:

$$B_{\mu n}(x, y) = \sum_I A^I_\mu(x)\,\omega^I_n(y) + (\text{massive Kaluza–Klein modes}). \tag{2.5}$$

One is effectively reducing B on one-cycles, one-dimensional submanifolds of the Calabi–Yau. But a Calabi–Yau manifold has no harmonic one-forms, $b_1(Y_6) = 0$, so these components vanish, and neither G_{MN} nor B_{MN} can give a vector $A_\mu(x)$ in four dimensions. We shall see later why this is a property of the Calabi–Yau.

The R–R forms offer more possibilities. C_1 in type IIA is already a one-form, and gives a vector directly. A form with more indices gives a vector if it is reduced on a cycle with one dimension less than its degree: C_3 in type IIA reduced on a two-cycle ($3 - 2 = 1$), or C_4 in type IIB reduced on a three-cycle ($4 - 3 = 1$). In principle all of these behave in four dimensions as objects with one index. This does not work either, for a reason that is a property of the effective theory: the R–R forms do not couple to any massless mode of the string. One can obtain some kind of photons from these forms, but the photons of the Standard Model must couple to electrons, to leptons and quarks, and the R–R photons couple only to massive objects, with masses at or above the string scale, which cannot be ordinary leptons or quarks: the very massive D-branes that we shall meet later. Moreover they are abelian. So one can obtain photons from the closed-string spectrum of type IIA and type IIB, but they are not the photon of the Standard Model.

Why there are no harmonic one-forms, and how many R–R photons there are. On a compact Ricci-flat manifold a harmonic one-form ω obeys the Weitzenböck identity $(\Delta\omega)_m = -\nabla^n\nabla_n\omega_m + R_m{}^n\omega_n$, whose last term vanishes. Contracting with ω^m and integrating by parts gives $\int_{Y_6} |\nabla\omega|^2 = 0$, so ω is covariantly constant (Bochner's argument). A covariantly constant vector is invariant under the holonomy group, but $SU(3)$ acts on the tangent space $\mathbb{R}^6 \simeq \mathbb{C}^3$ as $\mathbf{3} \oplus \bar{\mathbf{3}}$, which contains no singlet. Hence $b_1 = 0$ for holonomy exactly $SU(3)$. The same argument shows that on a compact Ricci-flat manifold every Killing vector is covariantly constant, so a Calabi–Yau has no

continued

continuous isometries either, and the Kaluza–Klein vectors $G_{\mu m}$ of Section 2.7 are absent too. The fundamental group of Y_6 is finite, but it can be non-trivial, as for the free quotients used in heterotic model building, where the non-contractible loops carry Wilson lines.
Counting the R–R vectors uses the Hodge numbers of Section 2.7. In type IIA, C_1 gives one vector and C_3, expanded in the $h^{1,1}$ harmonic two-forms, gives $h^{1,1}$ more. In type IIB, C_4 expanded in the $b_3 = 2h^{2,1} + 2$ harmonic three-forms gives $h^{2,1} + 1$ vectors, half of the naive number, because F_5 is self-dual and the electric and magnetic components are not independent. In four-dimensional $\mathcal{N} = 2$ language these are the graviphoton and the vectors of $h^{1,1}$ (IIA) or $h^{2,1}$ (IIB) vector multiplets, and the orientifold projection removes some of them. The charged objects are D-branes wrapped on cycles: D0-branes for C_1, D2-branes on two-cycles for C_3, D3-branes on three-cycles for C_4. A Dp-brane has tension $T_p = 2\pi M_s^{p+1}/g_s$ in our conventions, so a brane wrapped on a cycle of volume vol ℓ_s^p is a particle of mass 2π vol M_s/g_s, at or above the string scale in the regime $g_s \ll 1$ of Equation (1.7).

This is a very interesting prediction of string theory. On top of the ordinary photon there would be many other abelian gauge bosons, called *dark photons*: dark because they do not couple to ordinary matter, but still present, and coupled to it gravitationally. All these massless dark photons would be around simply as a consequence of string theory, and the question is how to detect them. One way, on which the lecturer has worked, uses cosmology. Whether the dark photons are present in the universe depends on whether they were produced cosmologically. If they were, they behave in practice like neutrinos: they have spin one rather than spin one half, but, like neutrinos, they carry no charge, and they are relativistic, because they are massless. They behave as a dark relativistic species, dark because of their very feeble interactions with the Standard Model, effectively decoupled, like extra neutrinos. They would be an extra energy density in the universe, which would modify its expansion rate. By measuring the expansion rate one can then either discover that there is something more than the three neutrinos, or argue that there cannot be more, which would be a strong constraint on these models.

Dark radiation and N_{eff}. The energy density of relativistic species after electron–positron annihilation is conventionally written as

$$\rho_{\text{rad}} = \left[1 + \frac{7}{8}\left(\frac{4}{11}\right)^{4/3} N_{\text{eff}}\right]\rho_\gamma,$$

where ρ_γ is the energy density of the photons and N_{eff} counts the neutrino-like species; the Standard Model predicts $N_{\text{eff}} \simeq 3.044$. Any additional dark radiation, such as relativistic dark photons (or the axions that we shall meet later), contributes $\Delta N_{\text{eff}} = \frac{8}{7}\left(\frac{11}{4}\right)^{4/3}\rho_{\text{dark}}/\rho_\gamma$. It changes the expansion rate during Big Bang nucleosynthesis and at recombination, and so the primordial abundances and the anisotropies of the cosmic microwave background. The Planck satellite measured $N_{\text{eff}} = 2.99 \pm 0.17$ [13], so present data allow ΔN_{eff} of a few tenths at most. A massless vector with two polarisations in thermal equilibrium with the neutrinos would give $\Delta N_{\text{eff}} = 8/7$ and is excluded; how much dark radiation a given compactification produces depends on how the dark sector is populated, for example by the decays of the moduli of Section 2.8.

2.5 Heterotic model building

For these reasons, in the mid-1980s people focused on the $E_8 \times E_8$ heterotic string. One E_8 factor is the *visible sector*: a symmetry breaking (a Higgs mechanism) inside it brings it down to a grand unified theory (GUT). The relevant subgroups are

$$E_8 \supset SO(10) \supset SU(5) \supset SU(3) \times SU(2) \times U(1). \tag{2.6}$$

$SO(10)$ and $SU(5)$ are the groups used for grand unification, and a Higgs mechanism, similar to the one that breaks $SU(2)_L \times U(1)_Y$ in the Standard Model, breaks $SU(5)$ down to $SU(3) \times SU(2) \times U(1)$. At the beginning there is a single group factor, so a single coupling: a unified theory with only one non-gravitational interaction, which then breaks down to the three interactions of the Standard Model.

The standard embedding and the number of generations. In a heterotic compactification on a Calabi–Yau threefold the internal components A^a_m of the gauge field cannot all vanish: the Bianchi identity of the three-form field strength, $\mathrm{d}H = \frac{\alpha'}{4}\big(\operatorname{tr} R \wedge R - \operatorname{Tr} F \wedge F\big)$ in a common normalisation, requires a gauge background on Y_6 to compensate for the curvature. The simplest solution, the *standard embedding* of Candelas, Horowitz, Strominger and Witten [9], identifies the gauge connection in an $SU(3)$ subgroup of one E_8 with the spin connection, whose holonomy is $SU(3)$. The unbroken group is the commutant of $SU(3)$ in E_8, i.e. E_6, a GUT group containing $SO(10)$: in four dimensions the chain Equation (2.6) starts as $E_8 \supset E_6 \supset SO(10)$, the first step being a breaking by the internal gauge background, which from the four-dimensional point of view acts like a Higgs mechanism. Under $SU(3) \times E_6$,

$$\mathbf{248} = (\mathbf{8}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{78}) \oplus (\mathbf{3}, \mathbf{27}) \oplus (\bar{\mathbf{3}}, \overline{\mathbf{27}}),$$

and the massless charged matter consists of $\mathbf{27}$'s and $\overline{\mathbf{27}}$'s, one of the Hodge numbers $h^{1,1}$, $h^{2,1}$ of Section 2.7 counting each. The net number of chiral generations is therefore $|h^{1,1} - h^{2,1}| = |\chi(Y_6)|/2$, half the Euler characteristic: a topological number. Each $\mathbf{27}$ contains a full Standard Model family, since $\mathbf{27} = \mathbf{16} \oplus \mathbf{10} \oplus \mathbf{1}$ under $SO(10)$. The quintic, with $\chi = -200$, gives 100 generations; the first three-generation models used a free $\mathbb{Z}_3$ quotient of the Tian–Yau manifold, with $\chi = -6$, whose non-contractible loops carry Wilson lines that break E_6 further.

The second E_8 factor is a *hidden sector*. "Hidden" means that the Standard Model particles, leptons and quarks, do not feel its interactions, since they are not charged under it, and similarly there is matter charged under the hidden E_8 that is not charged under the Standard Model. The hidden sector is responsible for supersymmetry breaking. In it there is a dynamics very similar to confinement in QCD. In QCD the gauge coupling grows towards low energies and blows up, and quarks and antiquarks condense: the quark bilinear develops a non-zero vacuum expectation value, $\langle q\bar{q}\rangle \neq 0$. Similarly, in the hidden E_8 there is strong dynamics at low energies, which gives a non-zero vacuum expectation value to some fields, and this breaks supersymmetry at low energies. Gravitational interactions, which are always present, then couple the hidden sector to the visible one and mediate the breaking to it, so that the visible sector also feels the breaking and becomes a non-supersymmetric Standard Model. That was the idea.

Gaugino condensation and gravity mediation. The mechanism behind "dynamically, via non-perturbative effects" is dimensional transmutation. In an asymptotically free gauge theory with one-loop coefficient $b_0 > 0$, i.e. $\mu\,\mathrm{d}g/\mathrm{d}\mu = -b_0 g^3/16\pi^2$, the coupling runs as

$$\frac{1}{g^2(\mu)} = \frac{1}{g^2(M)} - \frac{b_0}{8\pi^2}\ln\frac{M}{\mu}, \qquad \text{and blows up at} \qquad \Lambda = M\,\exp\!\Big(-\frac{8\pi^2}{b_0\, g^2(M)}\Big).$$

Λ is not analytic in g at $g = 0$, so it is invisible at every order of perturbation theory, and it is naturally exponentially smaller than M: with $g^2(M)/4\pi = 1/25$ and $b_0 = 9$ (pure $\mathcal{N} = 1$ super-Yang–Mills theory with gauge group $SU(3)$, $b_0 = 3N$), $\Lambda \simeq 3 \times 10^{-8} M$. In a hidden sector without matter, pure super-Yang–Mills theory, there are no quarks, and the analogue of $\langle q\bar{q}\rangle$ is the gaugino bilinear, $\langle\lambda\lambda\rangle \sim \Lambda^3$: *gaugino condensation*. In global supersymmetry the condensate alone does not break supersymmetry. In supergravity, however, the gauge coupling is the expectation value of a modulus (the dilaton, in the heterotic string), the condensate appears as a non-perturbative superpotential $W_{\mathrm{np}} \propto e^{-cf}$, with f the gauge kinetic function and $c > 0$ a constant fixed by the gauge group, of the type of Equation (2.1), and the resulting potential for

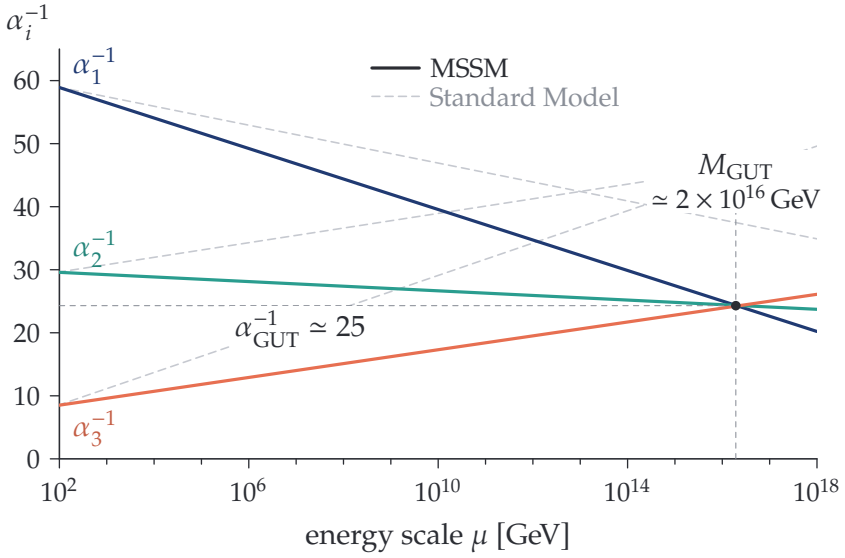


Figure 2.2: Running of the three inverse gauge couplings of the Standard Model gauge group with the MSSM spectrum, at one loop. They meet at $\alpha_{\text{GUT}}^{-1} \simeq 25$ at $M_{\text{GUT}} \simeq 2 \times 10^{16}$ GeV.

continued

the moduli can break supersymmetry through their F-terms. The order parameter is $F \sim \Lambda^3/M_P$, and gravity mediation then gives soft masses in the visible sector of order the gravitino mass,

$$m_{\text{soft}} \sim m_{3/2} \sim \frac{F}{M_P} \sim \frac{\Lambda^3}{M_P^2},$$

which is 1 TeV for $\Lambda \sim 2 \times 10^{13}$ GeV. Non-perturbative superpotentials of this type will be a key ingredient of KKLT and of the Large Volume Scenario.

2.6 The problems of the heterotic string: the GUT coupling

The heterotic string, however, has problems, which still have to be fully addressed, and people are still working on them. The first concerns the gauge coupling. In a heterotic compactification that leads to a grand unified theory, based on $SO(10)$ or $SU(5)$, one can compute the coupling of the GUT, and one finds that it is controlled by the string coupling and by the volume of the Calabi–Yau manifold in string units (for now, take it for granted; see the addition below):

$$\alpha_{\text{GUT}} \equiv \frac{g_{\text{GUT}}^2}{4\pi} \simeq \frac{g_s^2}{\mathcal{V}}. \tag{2.7}$$

Since our Einstein frame coincides with the string frame in the vacuum, $\mathcal{V}$ is the volume in either frame (see the conventions). From phenomenology, on the other hand, we know the value of α_{GUT}. The three non-gravitational couplings of the Standard Model, run with the renormalisation group of the MSSM, tend to unify at a value

$$\alpha_{\text{GUT}}^{-1} \simeq 25, \tag{2.8}$$

as shown in Figure 2.2. This is the value needed for phenomenology.

But then Equation (2.7) requires

$$\mathcal{V} \simeq 25\, g_s^2, \tag{2.9}$$

and this means that one cannot have $g_s \ll 1$ and $\mathcal{V} \gg 1$ at the same time. These were the two conditions, Equation (1.13), to trust the effective theory. If one goes to the limit in which the string coupling is small, the volume becomes of order one or smaller, and there are very large higher-derivative (α') corrections to the effective theory. Vice versa, if one keeps the volume relatively large, so as to suppress the α' corrections, the string coupling is large, the loop corrections are large, and they can never be ignored. Either way it is hard to trust the effective theory. We shall see that this is not the case in type II theories. There the gauge fields live on D-branes, and the relation between the GUT coupling and the overall volume does not hold: the gauge coupling is set by the size of the cycle that the branes wrap, not by $\mathcal{V}$, so one can really go to large volume and small string coupling while keeping $\alpha_{\text{GUT}}^{-1} \simeq 25$.

The heterotic gauge coupling. The heterotic gauge bosons are closed-string modes, so their kinetic term comes from the sphere, like the Einstein–Hilbert term of Equation (1.5), and it carries the same factor $e^{-2\phi}$:

$$S_{10} \supset \int \mathrm{d}^{10}X \, \sqrt{-G} \, e^{-2\phi} \Big(\frac{1}{2\kappa_{10}^2} R_{10} - \frac{1}{4g_{10}^2} F^a_{MN} F^{aMN} \Big),$$

where the ratio g_{10}^2/κ_{10}^2 is a pure number times $1/\alpha'$, fixed by string theory ($g_{10}^2 = 4\kappa_{10}^2/\alpha'$ in the normalisation of Polchinski [7]). Reducing on Y_6 exactly as in Equations (1.9) and (1.10), the zero mode $A_\mu^a(x)$, constant on Y_6, has a four-dimensional kinetic term with

$$\frac{1}{g_{\text{GUT}}^2} = \frac{e^{-2\phi_0} \, \text{Vol}(Y_6)}{g_{10}^2}, \qquad \text{while} \qquad M_P^2 = \frac{e^{-2\phi_0} \, \text{Vol}(Y_6)}{\kappa_{10}^2}.$$

Both are proportional to $\mathcal{V}/g_s^2$, which gives Equation (2.7), and their ratio does not depend on g_s or $\mathcal{V}$ at all: $g_{\text{GUT}}^2 M_P^2 = g_{10}^2/\kappa_{10}^2 \propto 1/\alpha'$. With Equation (1.11) this says that

$$\alpha_{\text{GUT}} \simeq \frac{g_s^2}{\mathcal{V}} \simeq 4\pi \, \frac{M_s^2}{M_P^2},$$

where the second step is exact for $\alpha_{\text{GUT}} = g_s^2/\mathcal{V}$, and the precise factor of order one depends on the normalisation of the gauge generators and, since the heterotic string has no D-branes to fix it, on that of g_s. In the heterotic string the string scale is therefore not free: for $\alpha_{\text{GUT}}^{-1} \simeq 25$, $M_s \sim M_P \sqrt{\alpha_{\text{GUT}}/4\pi} \sim 10^{17}\,\text{GeV}$, an order of magnitude above the scale $M_{\text{GUT}} \simeq 2 \times 10^{16}\,\text{GeV}$ at which the couplings of Figure 2.2 meet. This mismatch is a long-standing puzzle of heterotic unification. Moreover Equation (2.9) with $g_s \lesssim 1$ gives $\mathcal{V}^{1/6} \lesssim 25^{1/6} \approx 1.7$, so that $M_{KK} = M_s/\mathcal{V}^{1/6}$ is hardly below M_s, and the window of ten-dimensional supergravity of Figure 1.2 closes. Finally, in four-dimensional language the heterotic dilaton combines with the axion dual to $B_{\mu\nu}$ into a chiral field whose real part is proportional to $\mathcal{V}\, e^{-2\phi}$, and which is the gauge kinetic function: α_{GUT}^{-1} is the expectation value of a modulus.

2.7 Moduli

The second problem allows us to introduce a very important concept, which applies to every string theory, not only to the heterotic string: *moduli stabilisation*. To explain what it means, consider the dimensional reduction of the ten-dimensional metric $G_{MN}(X)$ to four dimensions. Starting from one metric in ten dimensions, one obtains many scalars in four dimensions, which are called *moduli*. Take, as in Equation (1.2), $X_{10} = \mathbb{R}^{1,3} \times Y_6$, with coordinates x^μ, $\mu = 0, 1, 2, 3$, on $\mathbb{R}^{1,3}$ and y^m, $m = 4, \ldots, 9$, on the Calabi–Yau manifold Y_6.

The metric $G_{MN}(x, y)$ splits into three kinds of components:

$$G_{MN}(x,y) \longrightarrow \begin{cases} G_{\mu\nu}(x), \\ G_{\mu m}(x,y) = \sum_I A^I_\mu(x)\, \omega^I_m(y) = 0 \quad \text{for a Calabi–Yau,} \\ G_{mn}(x,y) = \sum_I \phi^I(x)\, \omega^I_{mn}(y). \end{cases} \tag{2.10}$$

The first, with both indices along $\mathbb{R}^{1,3}$, gives the four-dimensional metric once the dimensional reduction is done and the y-dependence is no longer seen. The mixed components $G_{\mu m}$ are, from the four-dimensional point of view, vectors. To find their zero modes, the massless ones, one expands in eigenstates of the internal Laplacian, here the harmonic one-forms $\omega^I_m(y)$, objects with one index in the extra dimensions, exactly as for $B_{\mu n}$ in Equation (2.5). For a Calabi–Yau there are none: the harmonic one-forms are related to one-cycles, and the one-cycles of a Calabi–Yau can all be shrunk to zero (more precisely, $b_1 = 0$, Section 2.4). So the metric never gives vectors.

The internal components G_{mn} are, from the four-dimensional point of view, scalars, since m and n are just labels. They can be expanded similarly, as a sum over scalar fields $\phi^I(x)$ times objects $\omega^I_{mn}(y)$ with two indices in the extra dimensions. How many terms the sum has depends on how many such objects can be defined on Y_6, which is a topological property of Y_6, and in practice the index I takes hundreds of different values. A metric in ten dimensions therefore gives a metric in four dimensions plus many, many scalars. These scalars parametrise the internal metric: moving around in the space of the scalar fields changes the metric of Y_6. Some of the deformations change the size of the Calabi–Yau and others its shape, as anticipated at the end of Section 1.8. Mathematicians call parameters of this kind, scalar parameters on which the metric of the Calabi–Yau depends, moduli. From the point of view of physics they are scalar fields: a nice relation between physics and mathematics. One of them, for instance, is the volume $\mathcal{V}$ that appeared in α_{GUT}, Equation (2.7). As a field, $\mathcal{V}(x)$ is a function on Minkowski space, and its geometric meaning is that the Calabi–Yau attached at the point x can have a different volume from the one attached at another point.

There are, in general, three types of moduli.

Kähler moduli. The first type parametrises deformations of the size of Y_6; an example is the volume. They are denoted

$$T_i(x), \qquad i = 1, \ldots, h^{1,1}(Y_6), \tag{2.11}$$

where $h^{1,1}$, a *Hodge number*, is the number of linearly independent harmonic $(1,1)$-forms on Y_6: a topological property. A form ω is harmonic if the Laplacian of Y_6 annihilates it, $\Delta\omega = 0$. The name $(1,1)$ comes from the fact that Y_6 is a complex manifold. On a complex manifold one can use holomorphic and antiholomorphic coordinates z^i and $\bar{z}^{\bar{\imath}}$, as one can describe the plane either with the coordinates (x, y) or, as the complex plane, with $(z, \bar{z})$. A $(1,1)$-form has two indices, one holomorphic and one antiholomorphic, $\omega_{i\bar{\jmath}}\, \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^{\bar{\jmath}}$, and in general a (p, q)-form has p holomorphic and q antiholomorphic indices.

Complex-structure moduli. The second type changes the complex structure of Y_6, i.e. how complex coordinates are assigned to it. Without changing the volume, these deformations change the shape of Y_6. They are denoted

$$U_a(x), \qquad a = 1, \ldots, h^{2,1}(Y_6), \tag{2.12}$$

where the Hodge number $h^{2,1}$, another large number, counts the linearly independent harmonic $(2,1)$-forms on Y_6: it is the dimension of the space of such forms. Even if the construction cannot yet be fully appreciated, its beauty is that these mathematical, topological

properties of Y_6 correspond, from the point of view of physics, to new scalar fields in four dimensions.

The axio-dilaton. The third type comes not from the metric but from the other scalars of the massless spectrum of type IIB, Table 2.1: the dilaton ϕ and the R–R zero-form C_0, a scalar since it has no index. They combine into a complex scalar, the *axio-dilaton*

$$S = e^{-\phi} + i\, C_0. \tag{2.13}$$

All the moduli turn out to be complex: T_i and U_a, like S, are complex scalar fields. The real part of S is $e^{-\phi}$, so by Equation (1.6) its expectation value is the inverse string coupling,

$$\langle \mathrm{Re}\, S \rangle = \frac{1}{g_s} \gg 1, \tag{2.14}$$

which must be large for the effective theory to be trusted.

For an arbitrary Calabi–Yau manifold, both $h^{1,1}(Y_6)$ and $h^{2,1}(Y_6)$ can be of order 100. Starting from one metric in ten dimensions one can thus obtain hundreds of scalars in four dimensions. Calabi–Yau manifolds are named after Eugenio Calabi, an Italian-born mathematician, who conjectured their existence, and Shing-Tung Yau, a Chinese-born mathematician, who proved it; mathematicians together with theoretical physicists then started building explicit examples. Their number is still not known. The largest dataset built so far, that of Kreuzer and Skarke [12], has about half a billion examples, 473 800 776 (it lists the four-dimensional reflexive polytopes, each of which defines Calabi–Yau threefolds as hypersurfaces in a toric variety), and each of them is a good string compactification. There are examples in which both Hodge numbers are small, for instance with $h^{1,1} = 1$ and $h^{2,1}$ also small, which give only a few scalars, but most examples have many. From the point of view of the compactification they are all equally good.

The moduli of a Calabi–Yau metric. The expansion of G_{mn} in Equation (2.10) can be made precise. On a Calabi–Yau threefold the deformations δg_{mn} that preserve Ricci-flatness split into two kinds. Those of mixed type, $\delta g_{i\bar{\jmath}}$, give the harmonic $(1,1)$-forms $i\,\delta g_{i\bar{\jmath}}\, \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^{\bar{\jmath}}$; they deform the Kähler form $J = i\, g_{i\bar{\jmath}}\, \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^{\bar{\jmath}}$, and there are $h^{1,1}$ real ones. Those of pure type, $\delta g_{\bar{\imath}\bar{\jmath}}$, are related to the harmonic $(2,1)$-forms χ by contraction with the holomorphic three-form Ω, $\delta g_{\bar{\imath}\bar{\jmath}} \propto \bar{\Omega}_{\bar{\imath}}{}^{kl}\chi_{kl\bar{\jmath}}$; they deform the complex structure, and there are $h^{2,1}$ complex ones. In total the metric has $h^{1,1} + 2h^{2,1}$ real moduli. Expanding the Kähler form in a basis ω_i of harmonic $(1,1)$-forms, $J = \sum_{i=1}^{h^{1,1}} t^i \omega_i$ (in units of ℓ_s^2), the t^i measure the volumes of two-cycles, and the volume of Y_6 is the cubic polynomial

$$\mathcal{V} = \frac{1}{3!}\int_{Y_6} J \wedge J \wedge J = \frac{1}{6}\,\kappa_{ijk}\, t^i t^j t^k, \qquad \kappa_{ijk} = \int_{Y_6} \omega_i \wedge \omega_j \wedge \omega_k,$$

with the triple intersection numbers κ_{ijk} of Y_6; the volumes of the four-cycles are $\tau_i = \partial\mathcal{V}/\partial t^i = \frac{1}{2}\kappa_{ijk}t^j t^k$. The complex partners of these real fields come from the p-forms of Table 2.1 expanded in the same harmonic forms; in the type IIB orientifolds studied later the Kähler moduli turn out to be $T_i = \tau_i + i\int_{D_i} C_4$, with D_i the four-cycles. Which moduli survive the orientifold projection will be seen then.

continued

The Hodge numbers of a Calabi–Yau threefold with holonomy exactly $SU(3)$ are summarised by its Hodge diamond, whose entries $h^{p,q}$, $0 \leq p, q \leq 3$, are

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 0 & & 0 & & \\ & 0 & & h^{1,1} & & 0 & \\ 1 & & h^{2,1} & & h^{2,1} & & 1 \\ & 0 & & h^{1,1} & & 0 & \\ & & 0 & & 0 & & \\ & & & 1 & & & \end{array}$$

The Betti numbers are sums along its rows: $b_1 = 0$, $b_2 = h^{1,1}$ and $b_3 = 2 + 2h^{2,1}$, and the Euler characteristic is $\chi = 2(h^{1,1} - h^{2,1})$. The quintic hypersurface in $\mathbb{P}^4$ has $h^{1,1} = 1$ and $h^{2,1} = 101$. Mirror symmetry pairs Calabi–Yau manifolds with $h^{1,1}$ and $h^{2,1}$ exchanged.
Finally, the reason why all these fields are massless at leading order is Yau's theorem (Section 1.7): for every choice of complex structure and of Kähler class there is a Ricci-flat metric. The ten-dimensional vacuum equations are therefore solved for every value of the moduli, there is a continuous family of solutions, and no potential for the moduli is generated at this level.

2.8 The moduli problem

What happens to the physics if there are all these scalars in four dimensions? The answer is that the moduli mediate new interactions. The moduli come from the dimensional reduction of the metric, so they couple to ordinary matter as the metric does: since in ten dimensions they are the metric, they couple gravitationally to everything. This also explains why we have not seen them yet: they carry no charge under the Standard Model gauge group, so they do not feel QCD, electromagnetism or the weak interactions. They are scalars which are around and which interact only gravitationally with the other particles. But then the way we see the gravitational interaction between two massive particles is through the exchange of a graviton, and on top of it there can be the exchange of any of these scalars: a fermion can emit a modulus through a gravitational vertex, and the modulus can be absorbed by another fermion. This is a new gravitational interaction on top of the Newtonian one, and it modifies Newton's law of gravity.

How can this be avoided? One has to check experimentally down to which distances between two masses the inverse-square law $1/r^2$ of Newton has been verified. The answer is a few tens of micrometres, which is not very small at all: below that distance we do not know how gravity behaves, whereas at the LHC the other interactions are probed down to 10^{-18} m. The reason is that when two bodies are pushed closer to each other the other interactions become more important, and it is hard to shield them. What this means depends on the masses of the moduli. If they are massless, they mediate modifications of gravity at very large distances. If they are massive, the force they mediate has a finite range, just as the weak interactions are felt only inside nuclei because the W and Z bosons are massive; then the modifications of gravity appear only at short distances. These new forces are called *fifth forces*, and to avoid them a string compactification must give a mass to all the moduli, a mass big enough that the fifth forces have not been seen. However, in a naive compactification most of them, and at first glance all of them, are massless, and there is no source to give them a mass. This is true at tree level, so one has to include quantum corrections and see whether they develop a potential for the scalars that gives them a mass, big enough to avoid the fifth forces. This is the problem of *moduli stabilisation*. Stabilisation refers to the fact that a massless field has no potential, or a flat potential, so one does not know where the field is: it can have very big quantum fluctuations and is not stabilised at any point. If instead a potential is generated,

it has a minimum, at which the field is stabilised, on top of acquiring a mass.

In summary, *moduli are uncharged four-dimensional scalars with gravitational-strength interactions with ordinary matter, since they come from the metric, which at leading (tree-level) order in supergravity are massless.* Their potential is flat: the deformations of the size and of the shape of the extra dimensions cost no energy. This would be a phenomenological disaster, for two reasons.

2.8.1 Fifth forces

The first reason is the one just described: massless moduli would mediate Yukawa-like fifth forces, which are not observed. The four-dimensional Lagrangian contains, together with the mass term of a fermion ψ of ordinary matter, such as the electron, an interaction of Yukawa type with a modulus,

$$\mathcal{L}_4 \supset \frac{m_\psi}{M_P}\,\phi\,\bar\psi\psi, \tag{2.15}$$

where ϕ stands for any of the moduli, canonically normalised, and the coupling up to a coefficient of order one is shown. The suppression by the Planck mass is typical of gravitational interactions. Through this coupling the exchange of ϕ between two particles of ordinary matter, Figure 2.3, gives a fifth force, with a vertex m_ψ/M_P at each end. The observational limit for moduli that couple to ordinary matter with strength $1/M_P$ is that, to avoid these fifth forces, they must be heavier than about a meV:

$$m_\phi \gtrsim 1\,\text{meV}. \tag{2.16}$$

The Yukawa potential of a modulus. Write the coupling as $\beta\,(m_\psi/M_P)\,\phi\,\bar\psi\psi$ with β of order one; in the non-relativistic limit $\bar\psi\psi$ is the number density of the particles, so a particle of mass m_i is a source of ϕ with charge $g_i = \beta m_i/M_P$. Exchange of a scalar of mass m_ϕ between two such sources gives the attractive potential $-\frac{g_1 g_2}{4\pi}\frac{e^{-m_\phi r}}{r}$, which adds to the Newtonian one. Since $1/(4\pi M_P^2) = 2G_N$,

$$V(r) = -\frac{G_N m_1 m_2}{r}\left(1 + \alpha_\phi\, e^{-r/\lambda_\phi}\right), \qquad \alpha_\phi = 2\beta^2, \qquad \lambda_\phi = \frac{1}{m_\phi}. \tag{2.17}$$

A massless modulus, $\lambda_\phi = \infty$, would simply rescale Newton's constant by $1 + \alpha_\phi$, but differently for bodies of different composition as soon as β is not universal, violating the equivalence principle, and it would change the bending of light, to which a scalar does not contribute: both are tested to high precision, which excludes it. A massive modulus modifies gravity only at $r \lesssim \lambda_\phi$, with $\lambda_\phi = 0.197\,\text{mm} \times (1\,\text{meV}/m_\phi)$. Torsion-balance experiments exclude $\alpha_\phi = 1$ for ranges above about $40\,\mu$m [14], i.e. they require $m_\phi \gtrsim 5\,$meV for a coupling of gravitational strength; Equation (2.16) is the round order of magnitude. Note also that the moduli couple to all massive particles, through the trace of their energy–momentum tensor, which is why the coupling in Equation (2.15) is proportional to the mass m_ψ.

2.8.2 Lack of predictivity

The second reason may already be guessed. We have seen, in the previous lecture and in this one, that all the features of the effective theory in string theory are moduli-dependent. There are no parameters that one can fix by hand, except α', and once it is fixed (or traded for M_P, Equation (1.11)) everything else is derived dynamically, as the vacuum expectation value of a field. For example, the coupling α_{GUT} of Equation (2.7), written in terms of the moduli with

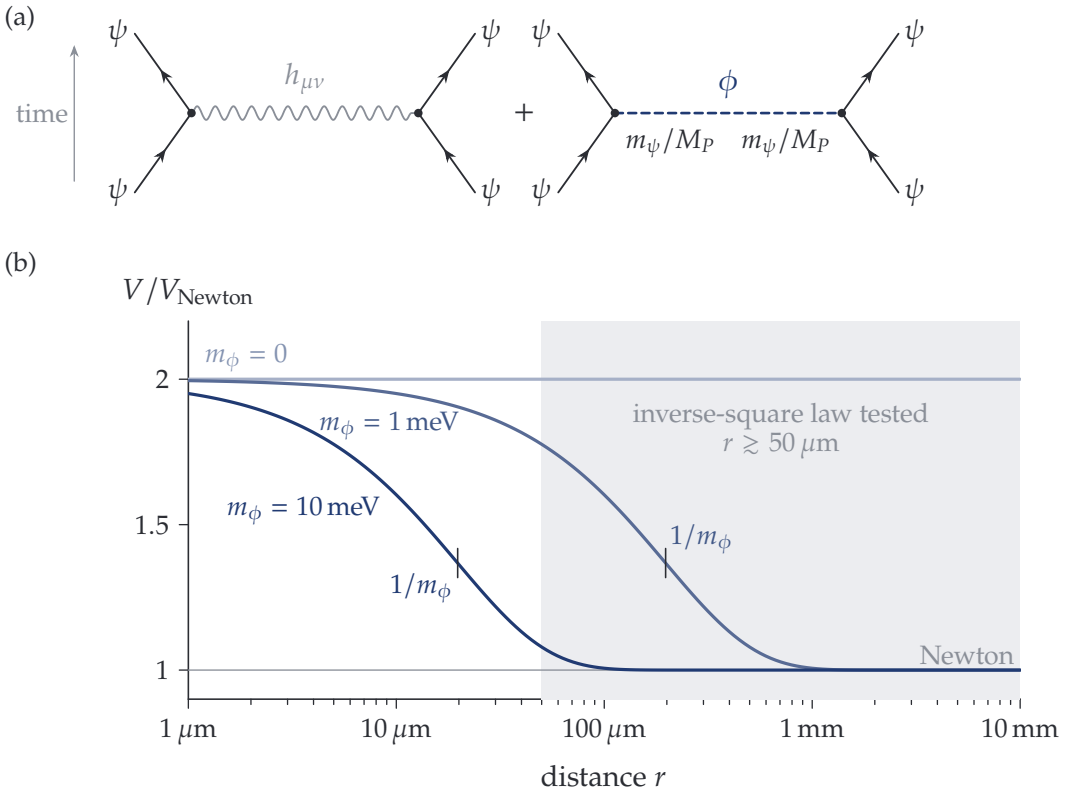


Figure 2.3: A modulus as the mediator of a fifth force. (a) Tree-level exchange of a modulus ϕ between two fermions, with the coupling of Equation (2.15) at each vertex, on top of the exchange of a graviton. (b) The resulting potential between two masses relative to Newton's, Equation (2.17), for a massless modulus and for two massive ones.

Equation (2.14), is

$$\alpha_{\text{GUT}} \simeq \frac{g_s^2}{\mathcal{V}} = \frac{1}{\langle \mathcal{V} \rangle \langle \text{Re}\, S \rangle^2}, \tag{2.18}$$

where $\langle \mathcal{V} \rangle$ is the expectation value of the volume, one of the moduli. So this parameter is not chosen by hand: it is the vacuum expectation value of a combination of two moduli. The point is that if these moduli are massless, their potential is flat, as said above. Then one cannot tell what the values of the fields are, because any value is allowed, with huge quantum fluctuations, and one cannot predict α_{GUT}, since neither $\mathcal{V}$ nor $\text{Re}\, S$ is known. The same holds for the scales of Chapter 1: the string scale, $M_s \simeq g_s M_P/\sqrt{\mathcal{V}}$ (Equation (1.11)), and the Kaluza–Klein scale, $M_{KK} \simeq g_s M_P/\mathcal{V}^{2/3}$ (Equation (1.15)), are different combinations of g_s and $\mathcal{V}$. Without knowing them one knows neither M_s, nor M_{KK}, nor α_{GUT}: one knows nothing.

There is therefore a huge lack of predictivity, because all the features of the four-dimensional low-energy effective theory of string compactifications are functions of the moduli. By all the features we mean the gauge couplings, for example the Yang–Mills coupling, the Yukawa couplings, the mass spectrum of the theory, the string scale, the Kaluza–Klein scale, the scale of supersymmetry breaking, the cosmological constant, the scale of cosmic inflation, and even the gauge group of the effective theory: all are functions of the moduli. This is the beauty of string theory, in a sense, because these quantities can be predicted rather than chosen by hand; but with massless moduli nothing is predicted.

> *Flat directions and predictions.* With an exactly flat potential every value ϕ_0 of a modulus defines a vacuum, and all these vacua have the same energy but different physics, different α_{GUT}, M_s, M_{KK} and so on. Nothing in the theory selects one of them, so choosing ϕ_0 is the same as choosing a parameter by hand. In cosmology the situation is worse: a massless field is not held anywhere, its value today is set by initial conditions and by its evolution, and during inflation it fluctuates by $\delta\phi \sim H/2\pi$ per Hubble time, with H the Hubble rate, so that its value need not even be the same in different regions of the universe. A potential with a minimum removes both problems: the field sits at the minimum, everything that depends on it becomes calculable, and the curvature of the potential there is the mass of the modulus, which must satisfy Equation (2.16).

What is needed is *moduli stabilisation*: the study of the quantum corrections to the scalar potential V (not to be confused with the volume $\mathcal{V}$), which can develop a minimum, stabilise the moduli ϕ^I there, make them massive and fix the predictions,

$$\left.\frac{\partial V}{\partial \phi^I}\right|_{\langle\phi\rangle} = 0, \qquad m_{IJ}^2 \propto \left.\frac{\partial^2 V}{\partial \phi^I \partial \phi^J}\right|_{\langle\phi\rangle} \text{ positive definite}, \tag{2.19}$$

with the mass matrix normalised by the kinetic terms of the moduli, and with masses large enough to satisfy Equation (2.16). These corrections will in the end generate minima, and then one can really tell where the moduli are. Next time we shall argue why moduli stabilisation is hard in the heterotic string. A hint can already be seen: we found in Section 2.6 that in the heterotic string it is hard to trust the effective theory, because the quantum corrections are hard to control, and yet the quantum corrections are exactly what is needed to stabilise the moduli.

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